Protection and Pricing Power in Sequential Venture Financing

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Abstract

Accelerator contracts are not negotiated, and they increasingly carry most-favored-nation (MFN) clauses under which the accelerator's uncapped SAFE converts on the best terms any investor obtains before the first priced round. Between the accelerator and that round sits an intermediate investor who covers an interim liquidity shock and chooses both a price and an amount; cash advanced beyond the shock survives to the round and lightens what the seed investor must contribute. The clause then has three pricing regimes with strictly ordered financing thresholds, and which one is reached is settled not by drafting but by how much the intermediate investor advances: one that merely bridges the shock absorbs the clause and leaves the threshold where the benchmark put it, while one that pre-funds the round pushes the clause into it and financing states are destroyed. The clause itself pushes toward the second, since it dilutes a small amount more than a large one. In every regime the intermediate investor's stake falls and the valuation it pays rises, so a higher interim valuation is no evidence that the reform helped; what sorts the regimes is the interim amount relative to the round it bridges to. The same protection is worth less to an investor that can price.

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1 Introduction

Accelerator contracts are not bargained. Y Combinator and Techstars run cohort-based programs of roughly three months that combine early financing with training, mentorship and investor networking, and the terms they offer are uniform across the cohort and imposed on follow-on investors (Zeisberger et al., 2017; Gonzalez-Uribe and Leatherbee, 2018). Until 2022, Y Combinator's standard deal was $125,000 for 7% of post-money equity. In January 2022 the accelerator added $375,000 invested on an *uncapped SAFE with a most-favored-nation provision*: a convertible instrument that automatically adopts the most favourable terms obtained by any other investor between the date of issuance and the next priced equity round.

The clause addresses a problem of sequence. The accelerator commits its capital at admission, before the cohort has produced any information, on terms it publishes in advance and applies uniformly to every company. Investors arriving later choose after observing what each company has become, and they negotiate individually. Nothing prevents such an investor from obtaining a more advantageous basis of calculation than the accelerator's own—a lower valuation, hence a larger share per dollar advanced—and the accelerator, having already invested and having renounced negotiation, has no instrument with which to respond. This is the commitment asymmetry that McAfee and Schwartz (1994) identify in sequential contracting, and the most-favored-nation clause is the standard remedy: it aligns the accelerator's conversion terms on whatever the intermediate investor obtains, so that moving first no longer means being treated worst.

Announcing the reform, Y Combinator's then-president argued that the larger amount would remove the immediate pressure to fundraise on unfavourable terms, encourage more founders to apply, and make it more likely that companies reach a successful seed round.

These are equilibrium claims about a multi-stage financing game, not accounting identities. Between the accelerator and the priced round, a company that runs short of cash must raise again, and the investor who covers that interim need—we call it the pre-seed investor—sets a price. The MFN clause makes the accelerator's effective stake a function of that price: the MFN SAFE converts at the lower of the pre-seed valuation and the round valuation. The clause therefore does not merely transfer equity; it changes what the pre-seed investor is solving for. Whether the extra capital improves financing outcomes depends on how that investor responds, and the response is not obvious in either direction.

A natural reading of the clause is that it is at worst neutral for the company: it moves equity between two investors and adds cash, so the firm should be weakly better off. That reading is right on one branch of the model and wrong on the other, and what separates them is not the clause but the size of the amount advanced under it.

Our model has four players and one state variable. An accelerator invests (σ_a, ρ_a)

before anything is known; a liquidity shock l then hits; a pre-seed investor may cover the residual need $b = l - \sigma_a$, choosing both how much to advance and at what valuation; the project's success probability π is revealed; and a seed investor funds whatever remains of the sunk cost I against whatever equity remains.

Five results follow. First, the pre-seed investor's equilibrium share is independent of the amount it advances: the valuation it sets scales one-for-one with the financing need, so across companies with similar characteristics but different liquidity shocks, pre-seed valuations should co-move with amounts raised while pre-seed ownership stakes should not.

Second, the amount is itself a choice, and a more consequential one than it appears. Cash advanced beyond the shock is not consumed by it; it sits on the balance sheet until the priced round, where it reduces what the seed investor must contribute. Pre-funding therefore buys a lower financing threshold, the trade-off is strictly convex, and the optimal amount is always a corner—either exactly the shock or the whole budget. What keeps the intermediate investor from simply buying the round is not its contract but the size of its balance sheet: with enough budget it pre-empts the seed round altogether, takes the entire residual equity, and the capitalization discipline that made it leave room for a successor disappears. Since an investor that funds the round is the round's investor rather than an intermediary, we bound its budget by the seed round's investment cost, which is what the label "pre-seed" asserts about the world and what keeps the seed investor's residual investment cost strictly positive. Within that bound, advancing exactly the shock—the case a first pass at this environment would write down—is one corner of a genuine choice, and the first result above is a statement about that corner; this paper states the condition under which it is the right one, and gives an observable proxy—the ratio of the interim amount to the round that follows—for deciding which corner the data sit in.

Third, and this is the paper's central mechanism, the pre-seed investor decides who bears the MFN SAFE, and the amount decides for it. Pricing low keeps the clause inside its own block; pricing high pushes it into the priced round, where the seed investor must clear it. Both branches are governed by the *same* threshold equation evaluated at two effective investment costs, J and $J + m$, and the two roots order as $\pi_1 < \pi_2$: passing the clause on raises the bar at the round. Two cutoffs in the amount separate three regimes—absorb, price exactly at the round, pass on—and which of them an investor can reach is settled by participation. The comparison that settles it is whether the residual cost it leaves the seed investor lies above or below $J^{\dagger}$, the point at which pre-funding stops paying. An investor that merely covers the shock is on the far side of that pivot and never leaves the first regime, where the clause is threshold-neutral: it raises the pre-seed valuation by exactly the amount it converts, transfers that equity to the accelerator, and leaves the seed threshold where the benchmark put it. One that pre-funds heavily crosses the pivot and reaches the other two, where the threshold rises to $\widehat{V}_p$ and then to π_2 and financing states are destroyed. The same cash that lowers the bar through J is what opens the regimes that raise it. Ordering the three regimes then adds a fact that belongs to none of them: at a given amount the pre-seed investor's stake is strictly below its benchmark value in all three, and the valuation at which it invests strictly above—one transfer read twice, since the price is the amount divided by the stake. Measured by the headline valuation, the interim round looks better under the clause in every regime, including those in which financing states are destroyed, so that measure tracks the transfer to the accelerator. What sorts the regimes is the ratio of the pre-seed valuation to the round that follows, together with the probability of reaching that round. The first regime leaves the threshold and the set of projects financed at it where the benchmark put them, and moves the capitalization table underneath: it is there that the intermediate investor gives up more than in either of the other two, since absorbing the clause whole is exactly what leaves the round untouched.

Fourth, and this is where the two margins meet, the clause acts on the amount in a way that carries it toward the pivot. Because the MFN SAFE removes a fixed sum from a block whose size the investor chooses, it dilutes a small amount far more than a large one, strengthens the incentive to pre-fund, and widens the set of budgets at which the investor pre-funds rather than merely bridging. That yields the paper's one unambiguous prediction about the reform—pre-seed rounds should get larger—but it also drives J down, and J falling below $J^{\dagger}$ is exactly what opens the regimes in which the clause stops being threshold-neutral. The instrument pushes the investor toward the region in which the instrument ceases to be harmless. The same pre-funding lowers the threshold within each regime, so the model does not sign the sum; it identifies the quantity that decides it, the interim amount relative to the round it bridges to.

Fifth, the clause is worth less to an investor who can price. Suppose the intermediate investor takes for itself the protection the accelerator holds against it, converting at the better of its base and the round price—the standard capped SAFE. Its problem then has two regimes rather than three, because above the cutoff the round price no longer depends on the base it names: the cap hands the round the conversion and the investor loses the ability to price at all. The instrument is weakly dominated by a plain fixed base, and strictly dominated exactly where the clause binds. A most-favored-nation clause protects an investor that cannot reprice and constrains one that can, so its value is a property of the holder rather than of the clause.

Taken together, the five say that the incidence of a contractual protection is decided by the balance sheet of the party it is written against, not by the protection. The question belongs to four literatures that have not met on this margin.

The first is the security-design literature on staged venture financing, which asks which instrument implements the efficient continuation decision when capital arrives in tranches. Gompers (1995) documents staging as the industry's central control device, and Neher (1999) shows that splitting the investment is what makes the entrepreneur's promise credible in the first place. Convertible securities are the recurring answer: Cornelli and Yosha (2003) obtain them from the entrepreneur's incentive to window-dress interim performance ahead of an abandonment option, Schmidt (2003) from double-sided moral hazard, Repullo and Suarez (2004) from a security-design problem over multiple stages, and Casamatta (2003) from the joint provision of capital and advice; the allocation of control that accompanies them is characterized by Hellmann (1998), and the learning-and-stopping structure by Bergemann and Hege (2005). In all of these the instrument is the unknown and the number of investors is two. Here the instrument is given by practice—a post-money SAFE, whose conversion basis is fixed at issuance—and the question is what a clause grafted onto it does once a third party prices an intermediate round.

The second is the literature on robust contracting and the dilution of early claims. Admati and Pfleiderer (1994) show that a fixed-fraction contract—the inside investor holds a constant share across rounds—is the unique contract that makes the insider's payoff immune to the manipulation of later-round pricing; the neutrality result below is its SAFE-era counterpart, in an instrument space where the early claim converts at a *price* rather than carrying a fixed share. What the contracts actually contain is documented by Kaplan and Strömberg (2003, 2004) and, for the link between investor characteristics and terms, by Bengtsson and Sensoy (2011), in the tradition of Sahlman (1990). That documentation predates the instrument studied here: Coyle and Green (2014) trace the displacement of priced seed rounds by convertible instruments, the contractual shift that made a standardized, non-negotiated pre-priced claim the normal entry point into a company's capital structure and thereby created the sequencing problem this paper studies.

The third is the industrial-organization literature on most-favored-customer clauses. McAfee and Schwartz (1994) show that nondiscrimination clauses protect the early signatories of a principal who contracts sequentially and cannot resist opportunism towards them, which is precisely the commitment asymmetry here: the accelerator invests before any uncertainty resolves, the pre-seed investor after observing the shock. That literature generally finds such clauses price-rigidifying and anticompetitive (Cooper, 1986; Besanko and Lyon, 1993), because the promise to match any better price removes the incentive to cut it. The sign reverses in this environment for a reason specific to the financing chain: within one regime the clause leaves the financing threshold of the chain untouched, and the party it constrains is not a seller facing rivals but an intermediary choosing where to place a claim it does not hold.

The fourth is the literature on the tiering of early-stage investors. Hellmann and Thiele (2015) study the equilibrium interaction between angel and venture capital markets, where a downstream tier holds up an upstream one—structurally the closest financing game to the sequence studied here, with the difference that their instruments are given whereas the instrument is the object of study in this paper. That tiers differ in what they bring, and that the difference is priced, is established by Hellmann and Puri (2002) and Hochberg et al. (2007); accelerators specifically are evaluated by Gonzalez-Uribe and Leatherbee (2018) and Hallen et al. (2020), who find effects on venture development but treat the program's terms as fixed. Two features of the environment make those terms worth modelling now: the collapse in the cost of starting a company has compressed the interval between the accelerator and the priced round (Ewens et al., 2018), and the risk that a company fails to raise its next round is a first-order determinant of early-stage financing (Nanda and Rhodes-Kropf, 2017; Lerner and Nanda, 2020). It is exactly that risk—the probability that the seed round does not happen—that the clause displaces between the parties.

The rest of the paper proceeds as follows. Section 2 lays out the environment—agents, valuations, and the timing of the three financing decisions. Section 3 solves a contract without the clause by backward induction and establishes the benchmark against which it is measured: the independence of the pre-seed share from the amount advanced, the pre-funding problem that decides the amount, and the partition of liquidity shocks into no-round, pre-seed and liquidation regions. Section 4 adds the MFN SAFE, separating its cash channel from its MFN channel, solving the pricing problem into three conversion regimes, showing which of them participation leaves reachable, and deriving what the clause does to the amount. Section 5 draws the empirical implications and returns to the claims made for Y Combinator's 2022 reform. Section 6 concludes, and all proofs are collected in the Appendix.

2 Model

2.1 Agents

Suppose there is no time preference and all the agents below are risk-neutral and profit-maximizing.

Let's introduce entrepreneurs without initial budget. Each entrepreneur carries a project requiring a two-stage financing path: surviving an interim liquidity shock $l \in (0, 1)$ and funding a sunk investment cost I at a priced seed round. Each project yields a stochastic outcome in $\{0, 1\}$. 0 means failure and 1 means success. Without l, the project cannot reach the seed round. Without I at the seed round, the project cannot succeed. A project is characterized by its probability of success $\pi \in [0, 1]$. Note that the probability of success is equal to the expected outcome. Each entrepreneur knows the expected outcome of his project. Let f be the probability density function of π, F denote its cumulative distribution function, g be the probability density function of l, G denote its cumulative distribution function.

Monopolistic accelerators and pre-seed investors are sponsors for entrepreneurs to help them survive the interim liquidity shock until the revelation of their projects' expected outcome and the seed stage. Accelerators invest ex-ante liquidity shock. Pre-seed investors invest ex-post liquidity shock.

Investment cost I must be financed by monopolistic seed investors after the revelation of the project's expected outcome π.

2.2 Allocations and valuations

When seed investors allocate σ_s to a project, it is with the promise that when payoffs are realized, they will take a *share* ρ_s of the realized outcome of the project. The *post-money valuation* V of a project corresponds to the ratio of the amount invested σ_s by the share ρ_s at the seed stage. The *pre-money valuation* is $V - \sigma_s$.

Without knowing either π or l but their respective distributions, the accelerator invests σ_a in exchange of a share ρ_a. After the liquidity shock l, the pre-seed investor invests σ_p in exchange of a share ρ_p. The share ρ_p can be fixed at the time of pre-seed investment or can be conditioned on the seed stage.

2.3 Definitions and assumptions

A project's *initial net present value* is defined as the difference of its expected outcome and the sum of the expected liquidity shock and the investment cost. For notational simplicity, the *initial market net present value*, which corresponds to the initial outside option's net present value, is taken to be 0.

(A1): $\mathbb{E}(\pi - l) - I > 0$ and f is log-concave

The first part of (A1) means that the average project is worth financing. This assumption can be thought of as conditioned on an accelerator preselection process supposed here independent from the game. The second part—that $\log f$ is concave—is a standard regularity condition satisfied by the uniform, normal, logistic, and exponential distributions, among others. It implies that the hazard rate $f/(1-F)$ is increasing.

Cash is fungible across dates. An amount advanced at the interim stage and not consumed by the shock stays on the balance sheet and is still there at the priced round, where it reduces the sum the seed investor must contribute. Nothing in the environment forces the pre-seed investor to advance exactly what the shock requires, and Section 3.3 shows that what bounds its amount is not the accounting but its own capacity.

Assumption 1 (B)**.** The pre-seed investor can advance at most β, with

$$\beta \leq I.$$

The accelerator's amount and the seed investor's contribution are unconstrained.

Two features of (B) matter, and both are descriptive rather than technical.

The constraint falls on the intermediate investor alone. Pre-seed vehicles are small relative to the rounds they precede, while the seed investor moves last, knows π, and is the marginal supplier of capital to a project whose value is by then established. The accelerator's budget is left free because its amount is a design object rather than a constraint.

The bound is the seed round's own investment cost. A pre-seed amount is measured in hundreds of thousands and the round that follows it in millions, so an intermediate investor able to fund I on its own is not an intermediate investor: it is the round. The seed round always happens, and the pre-seed investor always has a successor to accommodate. This is what makes the pricing problem of Sections 3–4 non-degenerate, and Remark 1 follows the argument to the corner a larger balance sheet would reach.

2.4 Timing

The game structure is as follows. First, without knowing either π or l but their respective distributions, the accelerator proposes (ρ_a, σ_a) to the entrepreneur. Then a liquidity shock occurs and the pre-seed investor proposes (ρ_p, σ_p), choosing an amount $\sigma_p \leq \beta$ that must at least cover the shock, and a valuation at which its claim converts. Whatever it advances beyond the shock stays in the firm. Finally, a market shock reveals π and enables a seed round proposal of $(\sigma_s, \rho_s) = (J, \frac{J}{V})$, where J is the part of the investment cost the firm has not already carried over.

The entrepreneur's outside option is normalized to zero at each stage.

Table 1 collects the notation. The primitives are the two shocks and the investment cost; everything else is either a contract—an amount and the share it buys—or an object the equilibrium generates.

3 Benchmark: fixed-equity SAFE

Before proceeding, it is useful to establish a benchmark result, namely that when the accelerator's contract is independent from the pre-seed and seed rounds, raising the amount invested increases seed success and the probability that no pre-seed round is needed. We solve by backward induction.

3.1 Seed stage

The seed investor moves last, knows π, and faces the capitalization table (ρ_a, ρ_p). It does not necessarily have to fund the whole of I: writing $\Delta \geq 0$ for the cash the pre-seed investor advanced beyond the shock and that the firm still holds, the seed investor's *residual investment cost* is

$$J := I - \Delta \in (0, I], \tag{1}$$

Panel A. Primitives and contracts	
π, f, F	Project's probability of success, its density and cdf
l, g, G	Interim liquidity shock, its density and cdf
I	Investment cost of the priced seed round
β	Pre-seed investor's budget, $\beta \leq I$ (Assumption 1)
(σ_a, ρ_a)	Accelerator's amount and share in the benchmark
(ρ_a^0, m)	Headline share, and the amount the accelerator places in an uncapped SAFE carrying an MFN clause, under the reform (Section 4)
(σ_p, ρ_p, V_p)	Pre-seed amount, share, and the price its claim converts at
(σ_s, ρ_s, V)	Seed amount, share, and post-money price of the round
$b = l - \sigma_a$	Residual need the accelerator's amount leaves
$\Delta = \sigma_p - b$	Cash advanced beyond the shock and still in the firm at the round
$J = I - \Delta$	Seed investor's residual investment cost, (1)
Panel B. Objects the equilibrium generates	
$M(x)$	$M(x) = \int_x^1 \pi f(\pi)\, d\pi$, the expected outcome contributed by projects above x; governs every participation decision
k	Equity base left to the investors arriving after the accelerator: $1 - \rho_a$ in Section 3, $1 - \rho_a^0$ from Section 4 on
$\pi_{\min}$, $\pi(J)$	Seed financing threshold, and the threshold map in the benchmark (Corollary 1)
φ_J, $W(J)$	Value of the convertible claim as a function of the threshold it engineers, and that value at its optimum, (2)
$\Lambda(J)$	$\Lambda(J) = M(\pi(J))/\pi(J)$, the marginal value of pre-funding a dollar of the round (Lemma 3)
$\tilde{\sigma}_p$	Amount cutoff separating exact funding from pre-funding (Proposition 3)
$\bar{x}$, $J^\dagger$	Pivot solving $M(\bar{x}) = \bar{x}$, and the residual cost at which $\pi(J^\dagger) = \bar{x}$, where Λ crosses one
$\bar{\sigma}_p$, $\bar{b}$	Participation caps on the amount and on the shock (Propositions 4 and 7)
ψ, D	$\psi = k\pi^3 f/D$, $D = \pi^2 f + M$; the threshold equation is $J = \psi(\pi)$, so $\pi(\cdot) = \psi^{-1}$ (Lemma 2)
$\Phi(\pi; J)$	Threshold equation both regimes collapse onto, $\Phi = D(\psi - J)$ (Definition 1)
π_1, π_2	Its roots at J and at $J + m$: the seed threshold when the clause converts in the pre-seed block, and when it converts into the round
$\widehat{V}_p$	Price at which the MFN SAFE is indifferent between converting at the pre-seed price and at the round price (Lemma 4)
T_1, T_2	Amounts at which the investor stops absorbing the clause and starts passing it to the round, (6)
$\bar{\sigma}_p^m$	Largest amount a participating investor advances under the clause (Proposition 10)
l_0, l_1, l_2	Shocks at which the figures cross a regime boundary: $l_0 = \sigma_a + m$, and $\sigma_p = T_1$, $\sigma_p = T_2$

Table 1: Notation. Panel A lists what is given before any decision is taken, together with the three contracts; Panel B lists the objects constructed from them, with the result that defines each. The only quantities carrying a decision are (σ_a, ρ_a), (σ_p, V_p) and the seed price V; Δ, J and b are accounting identities that follow from them.

strictly positive because Assumption (B) caps the pre-seed investor's amount at I, so that covering the shock leaves it strictly less than I to park. It solves

$$\max_V \int (\frac{J}{V(\pi)}\pi - J) f(\pi) d\pi$$

under: $\rho_a + \rho_p + \frac{J}{V(\pi)} \leq 1$

Write $\pi_{\min}$ for the lowest probability of success the seed investor is willing to finance.

Proposition 1. *The seed investor proposes the valuation $V = \frac{J}{1-\rho_a-\rho_p}$ and finances exactly the projects above $\pi_{\min} = V$.*

Some worth financing projects don't reach the seed stage. Two features of $\pi_{\min}$ drive everything below. It rises with the block of claims converting ahead of it, $\rho_a + \rho_p$—the familiar crowding channel—and it falls, one for one in the numerator, with any cash the firm brings to the round. The pre-seed investor controls both: the first through its price, the second through its amount.

3.2 Pre-seed stage

The pre-seed investor moves at $t = 4$, after the liquidity shock l has been realised but before π is revealed. It must cover the residual need $b := l - \sigma_a$, chooses the amount $\sigma_p \leq \beta$ it advances, and chooses the valuation V_p at which its claim converts—or declines to participate. Its share is $\rho_p = \sigma_p / V_p$, so choosing V_p is equivalent to choosing ρ_p.

Throughout, write

$$M(x) \; := \; \int_x^1 \pi \, f(\pi) \, d\pi, \qquad \text{so that} \qquad M'(x) \; = \; -x f(x).$$

$M(x)$ is the expected outcome contributed by projects above x; it is the object that governs every participation decision below.

The two halves of the pre-seed investor's decision—how much to advance, and at what price—separate cleanly, and only one of them is settled here.

Lemma 1. *Conditional on proposing, the pre-seed investor advances at least the residual need: $\sigma_p \geq b = l - \sigma_a$.*

Under-funding is dominated because it buys nothing: the shock is left uncovered, the firm is liquidated before π is revealed, the claim never converts, and the amount is sunk.

Over-funding is a different matter. An amount larger than the shock leaves $\Delta = \sigma_p - b$ on the balance sheet, and by Proposition 1 the seed investor's residual investment cost falls to $J = I - \Delta$. Fix Δ and carry $J = I - \Delta$ through the algebra. The pre-seed investor is repaid only on projects that reach and clear the seed round, that is on $\{\pi \geq \pi_{\min}\}$, and loses its outlay otherwise:

$$\max_{V_p} \int_{\pi_{\min}}^{1} \Big(\tfrac{\sigma_p}{V_p}\,\pi - \sigma_p\Big) f(\pi)\,d\pi \;-\; \int_0^{\pi_{\min}} \sigma_p\, f(\pi)\,d\pi \;=\; \max_{V_p} \underbrace{\sigma_p\left(\frac{M\big(\pi_{\min}(V_p)\big)}{V_p} - 1\right)}_{=:\,\Pi_p(V_p)}.$$

Proposition 2. *At a given amount, the pre-seed investor proposes the valuation V_p^* satisfying*

$$\sigma_p\, f(\pi_{\min})\,\pi_{\min}^3 \;=\; J\,V_p\,M(\pi_{\min}).$$

Write $\pi(J)$ for the threshold the pre-seed investor's optimal price engineers, as a function of the residual investment cost it leaves.

Corollary 1. *At a given amount, $\pi(J)$ solves*

$$(1-\rho_a)\, f(\pi)\,\pi^3 \;=\; J\Big(M(\pi) + f(\pi)\,\pi^2\Big),$$

an equation in which σ_p does not appear.

Consequently the equilibrium pre-seed share $\rho_p^* = 1 - \rho_a - J/\pi(J)$, the threshold $\pi(J)$, and the conditional seed probability $1 - F(\pi(J))$ depend on the amount only through the pre-funded part Δ, and not through the residual need b. In particular, if the amount covers exactly the shock ($\Delta = 0$, $J = I$), all three are independent of b and the valuation V_p^* scales one-for-one with b.

The economics is transparent: the monopolist pre-seed investor prices so as to hold a target share of the company—the share that optimally trades off its stake against the seed investor's financing constraint—and the valuation is merely the instrument through which any cash amount is converted into that share. The qualification matters, and it is what the next subsection exploits. Cash advanced to *cover* the shock is invisible to the threshold; cash advanced *beyond* it is not, because it changes J.

3.3 Pre-funding the seed round

Two objects recur from here to the end of the paper. Write k for the equity base the capitalization table leaves to the investors arriving after the accelerator—so $k := 1-\rho_a$ in this section, and $k := 1 - \rho_a^0$ from Section 4 on, where the accelerator holds a headline share plus an MFN SAFE—and let

$$\varphi_J(\pi) \;:=\; \Big(k - \frac{J}{\pi}\Big) M(\pi), \qquad W(J) \;:=\; \max_{\pi} \varphi_J(\pi) \;=\; \varphi_J\big(\pi(J)\big) \tag{2}$$

be the expected value of the convertible claim as a function of the threshold it engineers, and that value at its optimum. The two are tied to the first-order condition already in hand: differentiating (2) and using $M'(\pi) = -\pi f(\pi)$ gives

$$\varphi_J'(\pi) \;=\; -\,\frac{k\,\pi^3 f(\pi) - J\big(\pi^2 f(\pi) + M(\pi)\big)}{\pi^2},$$

so the threshold equation of Corollary 1 is exactly the first-order condition of that maximization—and, by Lemma 2 below, $\pi(J)$ is its maximizer rather than merely a critical point. Both objects therefore inherit whatever base the section they are read in supplies; only k changes when the accelerator's contract does. The pre-seed investor's maximized payoff at a given amount is then

$$\Pi_p \;=\; W(J) \;-\; \sigma_p. \tag{3}$$

It captures the whole claim it has engineered and pays for it with the amount it advances. Writing that amount as $\sigma_p = b + \Delta$ and the residual investment cost it leaves the seed investor as $J = I - \Delta$, the pre-seed investor's remaining problem—having already priced optimally—is

$$\max_{\Delta \in [0,\, \beta - b]} \;\; \Pi_p(\Delta) \;=\; W(I - \Delta) \;-\; b \;-\; \Delta, \tag{4}$$

the range coming from Assumption (B) net of what the shock already required (Lemma 1): covering the shock is prior, and pre-funding is a decision about the residual budget $\beta - b$ alone. An investor with $\beta < b$ has no such decision to make—it does not participate.

The threshold map $\pi(\cdot)$ and the value W are built from a single ratio. Write

$$D(\pi) \;:=\; \pi^2 f(\pi) + M(\pi), \qquad \psi(\pi) \;:=\; \frac{k\,\pi^3 f(\pi)}{D(\pi)}, \qquad \pi \in (0,1), \tag{5}$$

so that two identities hold by inspection: the threshold equation of Corollary 1 reads $J = \psi(\pi)$, and the derivative computed after (2) is $\varphi_J'(\pi) = -D(\pi)\big(\psi(\pi) - J\big)\big/\pi^2$. The notation $\pi(J)$ presumes three further things: that the equation has a unique solution, that this solution maximizes φ_J, and that it rises with J. All three follow from a single property of ψ.

Lemma 2 (The threshold map)**.** *Under (A1), ψ is a strictly increasing bijection from $(0,1)$ onto $(0,k)$, and $\psi(\pi) < k\pi$. Hence, for every $J \in (0,k)$:*

(i) the threshold equation has a unique solution $\pi(J) = \psi^{-1}(J)$, and $\pi(J) > J/k$;

(ii) $\pi(J)$ maximizes φ_J, which strictly increases below it and strictly decreases above it, so $W(J) = \varphi_J\big(\pi(J)\big)$;

(iii) $\pi(\cdot)$ is strictly increasing.

Write $\Lambda(J) := M(\pi(J))/\pi(J)$ for the value of one point of equity at the threshold the claim engineers, divided by the price at which that point is bought.

Lemma 3 (Convexity)**.** *$W'(J) = -\Lambda(J)$, and under (A1) Λ is strictly decreasing, so*

$$\Pi_p'(\Delta) \;=\; \Lambda(I-\Delta) \;-\; 1$$

is strictly increasing in Δ: the objective in (4) *is strictly convex.*

The interpretation is direct. Recall that the seed investor prices at $V = \pi_{\min}$ (Proposition 1), so it takes $J/\pi_{\min}$ of the company to fund J: one unit of investment cost lifted off its plate frees $1/\pi_{\min}$ points of equity for the pre-seed investor, and each point is worth $M(\pi_{\min})$. At the equilibrium threshold $\pi_{\min} = \pi(J)$, pre-funding one unit therefore returns $M(\pi(J))/\pi(J) = \Lambda(J)$ and costs exactly one. Now pre-funding lowers that threshold: $\pi(J)$ is strictly increasing in J, since a seed investor with less to contribute needs less equity to break even and will fund weaker projects—this is Lemma 2, and it holds after the pre-seed investor has re-optimized its price. Convexity follows because both factors of the return move the same way as the threshold falls: $1/\pi(J)$ rises, since at a lower price the seed investor takes more equity per unit it contributes, so removing one unit frees more of it; and $M(\pi(J))$ rises, since more projects survive to convert, so each point freed is worth more. The next unit advanced is worth strictly more than the last. This settles the shape of the answer before its location—a convex objective on an interval is maximized at an endpoint, so $\sigma_p^* \in \{b, \beta\}$ regardless of parameters. What remains is to say which.

Write $\tilde{\sigma}_p = b + \tilde{\Delta}$ for the amount at which the two candidates of (4) are worth the same.

Proposition 3 (A single threshold)**.** *The pivot $\tilde{\sigma}_p$ exists, is unique in $[b,\, b+I]$, and for every budget β admissible under (B),*

$$\sigma_p^* \;=\; \begin{cases} b & \text{if } \beta \le \tilde{\sigma}_p, \\ \beta & \text{if } \beta > \tilde{\sigma}_p. \end{cases}$$

Moreover $\tilde{\Delta} \ge 0$ is the same number for every residual need b.

Covering the shock therefore shifts the whole schedule, leaving unchanged how much pre-funding beyond it is worth doing.

The appendix constructs $\tilde{\Delta}$ from two objects reused in Section 4: the pivot $\bar{x}$, the unique solution of $M(\bar{x}) = \bar{x}$, and the residual investment cost $J^\dagger$ at which $\pi(J^\dagger) = \bar{x}$—where Λ crosses one. These locate the minimizer of Π_p at $\Delta^\dagger := I - J^\dagger$; the payoff falls to that point and rises past it, leaving the two endpoints of $[0, \beta - b]$ as the only candidates, and the comparison between them is monotonic in β.

Two consequences carry into the rest of the paper. First, on the pre-funding branch the seed investor's residual investment cost is $J = I - \beta + b$, so the financing threshold

$\pi(J)$ is strictly increasing in the shock b and strictly decreasing in the budget β: a deeper pocket upstream lowers the bar at the priced round, a larger interim shock raises it. On the other branch it is $\pi(I)$, constant in both. This is the comparative static the cash channel of Section 4 runs on. Second, Assumption (B) keeps $J > 0$ on either branch, so the seed round occurs in every financed state. Pre-funding acts on the seed constraint alone: reaching the priced round still requires the shock to have been absorbed, and no amount can buy its way past that—the liquidity constraint is what its first b units are for.

Remark 1 (Why a monopolist leaves room)**.** A monopolist pre-seed investor leaves room because it needs a successor: every point of share it takes is a point the seed investor cannot have, and that investor will not fund J out of a stub worth less than J. Pre-funding erodes that discipline, and (B) bounds the erosion. Relax the bound to $\beta \geq b + I$ and the argument runs to its corner—the investor advances $b + I$, the residual investment cost and the threshold fall to zero, and it holds the entire residual equity $1 - \rho_a$ for a payoff $(1 - \rho_a)\mathbb{E}(\pi) - b - I$ (appendix). It then prices against no one, which makes it a different institution: an investor that funds the round is the round's investor. What keeps an intermediary intermediate is the size of its balance sheet rather than the terms of its contract.

3.4 Which firms need a pre-seed round

Proposition 3 leaves the pre-seed investor in one of two regimes, and both are worth writing down. Call the investor *constrained* when $\beta \leq \tilde{\sigma}_p$, so that it advances the shock and no more—$\sigma_p = b$, $\Delta = 0$, $J = I$—and *pre-funding* when $\beta > \tilde{\sigma}_p$, so that it advances its whole budget and $\sigma_p = \beta$, $J(b) = I - \beta + b$. The partition of shocks differs between them in exactly one respect: whether the financing threshold is constant across the region where a pre-seed round occurs. We take the constrained case first, writing $\pi^*_{\min} := \pi(I)$ for the threshold it delivers.

Write

$$\bar{\sigma}_p \;:=\; M(\pi^*_{\min})\,\rho^*_p \;=\; M(\pi^*_{\min})\left(1 - \rho_a - \frac{I}{\pi^*_{\min}}\right)$$

for the expected value of the claim the pre-seed investor engineers at its optimal price.

Proposition 4. *The pre-seed investor participates if and only if $\sigma_p = l - \sigma_a \leq \bar{\sigma}_p$.*

The cap is $\bar{\sigma}_p = W(I)$: by (3) the most cash the pre-seed investor can sink is the expected value of the claim it engineers. There is therefore a maximum liquidity shock, determined entirely by (ρ_a, I, F), beyond which no pre-seed financing occurs at any price. Beyond it no valuation works—lowering V_p to grab a larger share raises $\pi_{\min}$ and destroys the very conversion states the share pays in.

Write $\pi_0 := I/(1 - \rho_a)$ for the threshold a firm faces when it reaches the seed round without a pre-seed investor on its cap table.

Proposition 5. *The realisation of l sorts every firm into one of three regimes:*

(N) No pre-seed round, *$l \leq \sigma_a$: here $(\rho_p, \sigma_p) = (0, 0)$, the seed investor finances the projects $\pi \geq \pi_0$, and the conditional seed probability is $1 - F(\pi_0)$.*

(P) Pre-seed round, *$\sigma_a < l \leq \sigma_a + \bar{\sigma}_p$: here $\rho_p = 1 - \rho_a - I/\pi^*_{\min}$, the seed investor finances the projects $\pi \geq \pi^*_{\min}$, and the conditional seed probability is $1 - F(\pi^*_{\min})$.*

(L) Liquidation, *$l > \sigma_a + \bar{\sigma}_p$: the shock exceeds what any investor will advance, the firm is liquidated at zero before the seed stage, and the conditional seed probability is 0.*

Moreover $\pi_0 < \pi^*_{\min}$.

The ordering is the price of survival: a pre-seed round costs equity, and the dilution it costs raises the bar at the seed round.

The two boundaries have different natures, which matters for the policy analysis below. The lower boundary σ_a is *mechanical*—a cash amount chosen before anything is known, moved one for one by the accelerator's amount. The upper boundary $\sigma_a + \bar{\sigma}_p$ is *endogenous*—$\bar{\sigma}_p$ is an equilibrium object, built from $\pi^*_{\min}$ and ρ^*_p, and it therefore moves with anything that shifts the pre-seed investor's problem, including the MFN SAFE of Section 4.

Proposition 6. *The unconditional probability of seed success is*

$$\mathbb{P}(\textit{seed}) \;=\; G(\sigma_a)\big(1 - F(\pi_0)\big) \;+\; \big(G(\sigma_a + \bar{\sigma}_p) - G(\sigma_a)\big)\big(1 - F(\pi^*_{\min})\big).$$

The pre-funding regime keeps this architecture and changes one thing: the threshold now moves with the shock across the region where a round occurs.

Suppose $\beta > \tilde{\sigma}_p$, so that the investor advances $\sigma_p = \beta$ wherever it participates, leaving the seed investor $J(b) := I - \beta + b$ against a residual need $b = l - \sigma_a$. By (3) its profit at that amount is $W\big(J(b)\big) - \beta$, so write b^W for the residual need at which an investor advancing its whole budget breaks even—the solution of $W\big(J(b^W)\big) = \beta$, taken to be β where the condition never binds—and set $\bar{b} := \min\{\beta,\, b^W\}$.

Proposition 7 (Pre-funding investors)**.** *On that branch:*

(i) a pre-seed round occurs if and only if the investor can cover the residual need and break even on it, $b \leq \beta$ and $W\big(J(b)\big) \geq \beta$; equivalently, if and only if $\sigma_a < l \leq \sigma_a + \bar{b}$;

(ii) the financing threshold is then $\pi\big(J(b)\big)$, strictly increasing in the shock;

(iii) the probability of seed success is

$$\mathbb{P}(seed) \;=\; G(\sigma_a)\big(1 - F(\pi_0)\big) \;+\; \int_{\sigma_a}^{\sigma_a+\bar{b}} \Big(1 - F\big(\pi(J(l-\sigma_a))\big)\Big)\, dG(l).$$

Two differences with Proposition 6 are worth naming. The product becomes an integral, because the shock distribution and the project distribution no longer separate: each shock now carries its own threshold. And the ordering $\pi_0 < \pi^*_{\min}$ can reverse, since a sufficiently pre-funded round leaves the seed investor less to contribute than no shock at all would have—a firm that suffers a small shock and meets a deep-pocketed investor ends up better placed than one that suffered none. That reversal is the subject of the qualification to Proposition 8 below.

This delivers the benchmark result announced at the outset of the section.

Proposition 8. *For a constrained pre-seed investor, increasing σ_a raises the probability of seed success, and raises the probability that no pre-seed round is needed.*

Both boundaries of Proposition 5 shift rightwards by the same amount, so firms move from the pre-seed region into the no-pre-seed region and out of liquidation, while π_0, $\pi^*_{\min}$, ρ^*_p and $\bar{\sigma}_p$ are all unchanged—they depend on (ρ_a, I, F) only. The accelerator's cash therefore works on the *extensive* margin, which firms need a pre-seed round at all, and not through the terms of that round.

That first half rests entirely on $\pi_0 < \pi^*_{\min}$: moving a firm out of the pre-seed region helps only if that region is the worse place to be. Once the investor pre-funds, the ordering is no longer automatic. On the pre-funding branch of Proposition 3 the threshold there is $\pi(J)$ with $J = I - \beta + b$, and since $\pi(\cdot)$ is increasing, $\pi(J) < \pi_0$ exactly when $J < J_0$, where J_0 is the residual investment cost the threshold equation associates with π_0. Under (B) this requires $b < J_0$: only a firm whose residual need is smaller than J_0, and whose investor spends its entire budget covering it and pre-funding the rest, ends up better placed than it would have been with no shock at all. The bound confines the reversal to a corner of the shock space rather than eliminating it. The second half—fewer firms need an interim round—holds regardless.

Figure 1 plots the whole equilibrium against the shock itself, holding every other parameter fixed: the three amounts, the three stakes, the two prices and the seed probability, each as a function of l. The partition is what one reads off the vertical lines—the pre-seed stake flat across the constrained region, and the seed probability a step function whose jumps are the boundaries the accelerator's cash moves.

Table 2 collects the equilibrium of this section. Two primitives sort the firm: the liquidity shock l, which decides whether a pre-seed round is needed and whether it can be financed at all, and the pre-seed investor's budget β, which decides how it is financed. Everything else in the table is a function of (ρ_a, I, F) and of those two. The two stakes are not independent entries: the seed investor prices to saturate the cap

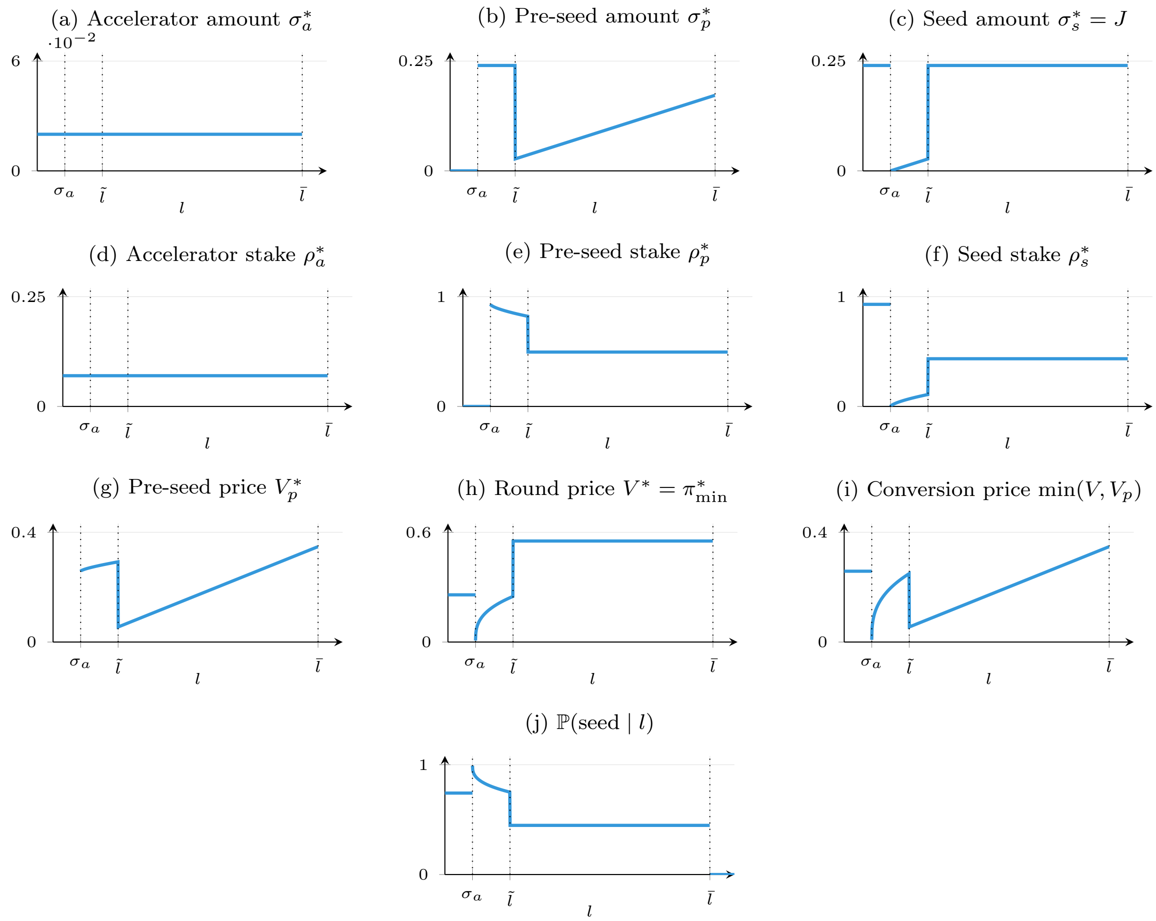

Figure 1: Equilibrium outcomes as functions of the liquidity shock l: the benchmark. Every other parameter is held fixed, so each panel is a cross-section of firms differing only in the shock they draw. Up to σ_a the accelerator's amount absorbs the shock and no pre-seed round takes place. Past it the investor enters and pre-funds—it advances its whole budget in (b), which is why the seed investor's residual cost in (c) starts near zero and climbs. At $\tilde{l}$ the shock has eaten the pre-funding: from there the amount merely covers the shock, J is back at I, and panels (e), (f), (h) and (i) go flat. That flat stretch is the share invariance of Corollary 1, and it is what identifies the regime in data; the price in (g) keeps rising one-for-one with the amount raised. The accelerator's two panels are flat throughout: its contract is signed before the shock and nothing revises it. At $\bar{l}$ the shock exceeds what the investor can or will advance and the firm is liquidated. Drawn for f uniform on $[0, 1]$, $\rho_a = \rho_a^0 = 0.07$, $I = 0.24$, $\sigma_a = 0.02$, $\beta = 0.24$—the budget is the largest Assumption (B) allows, $\beta = I$, which is what puts all three conversion regimes within reach: a smaller one confines the investor to the first. The parameters are illustrative: the levels of the stakes reflect the monopolistic closure, and it is their shape in l that the figure is about.

table, so $\rho_s^* = J/\pi(J)$ and the pre-seed stake is what is left of $1-\rho_a$ once that is taken. Reading a column down therefore shows the same primitive twice—what the pre-seed investor takes is what the seed investor is left unable to claim. The accelerator's two rows are constant across the table: its contract is signed before anything is known and nothing in the two rounds that follow revises it. That is exactly what Section 4 changes.

	(N) no pre-seed $l \le \sigma_a$	(P) constrained $\beta \le \tilde{\sigma}_p$	(P) pre-funding $\beta > \tilde{\sigma}_p$	(L) liquidation l beyond the cap
Accelerator σ_a^*	σ_a	σ_a	σ_a	σ_a
Accelerator ρ_a^*	ρ_a	ρ_a	ρ_a	ρ_a
Pre-seed σ_p^*	0	$l - \sigma_a$	β	—
Pre-seed ρ_p^*	0	$1 - \rho_a - \dfrac{I}{\pi(I)}$	$1 - \rho_a - \dfrac{J}{\pi(J)}$	—
Seed $\sigma_s^* = J$	I	I	$I - \beta + l - \sigma_a$	—
Seed ρ_s^*	$1 - \rho_a$	$\dfrac{I}{\pi(I)}$	$\dfrac{J}{\pi(J)}$	—
Pre-seed price V_p^*	—	$\dfrac{l - \sigma_a}{1 - \rho_a - I/\pi(I)}$	$\dfrac{\beta}{1 - \rho_a - J/\pi(J)}$	—
Round price $V^* = \pi^*_{\min}$	$\dfrac{I}{1 - \rho_a}$	$\pi(I)$	$\pi(J)$	—
$\mathbb{P}(\text{seed} \mid \cdot)$	$1 - F\big(\frac{I}{1-\rho_a}\big)$	$1 - F\big(\pi(I)\big)$	$1 - F\big(\pi(J)\big)$	0

Table 2: Equilibrium of the benchmark, by regime. Every entry is a function of the primitives $(\rho_a, \sigma_a, I, \beta, l)$ and of the distributions, through the threshold map $\pi(\cdot)$ of Corollary 1—the root of $\Phi(\cdot\,; J)$, which depends on ρ_a, on the residual investment cost, and on f. In the pre-funding column the pre-seed investor advances β, of which $\beta - (l - \sigma_a)$ is left in the firm, so the seed investor's amount is $\sigma_s^* = J = I - \beta + l - \sigma_a$; in the two other financed columns $\sigma_s^* = J = I$. The remaining rows are written in J rather than in that expression, since J is the argument of the threshold map $\pi(\cdot)$ throughout the paper. The two (P) columns share the shock conditions $\sigma_a < l \le \sigma_a + \bar{\sigma}_p$ and $\sigma_a < l \le \sigma_a + \bar{b}$ respectively and are separated by the budget cutoff $\tilde{\sigma}_p$ of Proposition 3; $\bar{\sigma}_p = W(I)$ and $\bar{b} = \min\{\beta, b^W\}$ are the participation caps of Propositions 4 and 7. The cap table saturates (Proposition 1), so $\rho_a + \rho_p^* + \rho_s^* = 1$ in every financed column, and the round is priced at the marginal project, so $V^* = \pi^*_{\min}$. The threshold is constant across the constrained column and rising in the shock in the pre-funding one, which is the single difference between them.

4 The accelerator's MFN SAFE

Suppose the accelerator now invests σ_a against a headline share ρ_a^0 *and*, in addition, an amount m whose conversion price is

$$\min\big(V,\, V_p\big),$$

the more favourable of the seed price and the pre-seed price. That amount is an *uncapped SAFE carrying a most-favored-nation clause*—uncapped because it names no base of its own, and we call it the accelerator's *MFN SAFE* for short. It automatically adopts the best terms granted to any investor entering before the first priced round, and in this environment there is exactly one such investor—the one that covers the interim shock.

The expression $\min(V, V_p)$ will reappear in Remark 2 attached to a different contract, so it is worth fixing what an MFN clause is. It carries *no base of its own*: it converts at the best price obtained by any investor entering after its holder. Here the accelerator has two such investors, at prices V_p and V, and neither is a number it chose—which is what makes the clause a remedy for having committed first. A valuation cap, by contrast, adds a base the holder specifies to that minimum. The two coincide algebraically whenever the holder has a single successor, as the pre-seed investor does, and Remark 2 shows what that coincidence is worth here.

Such a contract has two distinct channels, which we treat in turn: a *cash* channel, since the accelerator's committed amount rises, and an *MFN* channel, since the term $m/\min(V, V_p)$ enters the capitalization table. The two need not push in the same direction, and separating them is the purpose of what follows.

4.1 The cash channel

The first channel is not new: it is Proposition 8, established in the benchmark. Raising σ_a moves the two boundaries of Proposition 5 rightwards, enlarging the no-pre-seed region and shrinking the liquidation region, and it does so without touching the terms of the pre-seed round—the equilibrium stake ρ_p^* and threshold $\pi^*_{\min}$ are the same numbers before and after. The cash component therefore acts on the *extensive* margin alone, and favourably.

4.2 The MFN channel

At $t = 7$ the seed investor now solves

$$\max_V \int \Big(\frac{J}{V(\pi)}\pi - J\Big) f(\pi)\, d\pi \qquad \text{subject to} \qquad \rho_a^0 + \frac{m}{\min(V, V_p)} + \rho_p + \frac{J}{V} \;\le\; 1.$$

As in the benchmark the objective is strictly increasing in the seed share, so the cap table binds. Which of the two prices the MFN SAFE converts at determines the shape of the constraint, and hence the equilibrium valuation:

$$V \;=\; \begin{cases} \dfrac{J+m}{1-\rho_a^0-\dfrac{\sigma_p}{V_p}} & \text{if } V<V_p \quad \text{(it converts at the round price)}, \\[2ex] \dfrac{J}{1-\rho_a^0-\dfrac{\sigma_p+m}{V_p}} & \text{if } V>V_p \quad \text{(it converts at the pre-seed price)}. \end{cases}$$

Write

$$\widehat{V}_p \;:=\; \frac{J+m+\sigma_p}{1-\rho_a^0}$$

for the pre-seed price at which the two branches meet.

Lemma 4. *$V<V_p$ if and only if $V_p>\widehat{V}_p$, and $V>V_p$ if and only if $V_p<\widehat{V}_p$.*

The economics is worth stating plainly. A pre-seed investor that prices the company *low* ($V_p<\widehat{V}_p$) hands the accelerator's MFN SAFE a cheap conversion: the MFN SAFE converts at V_p and lands in the pre-seed block of the cap table. A pre-seed investor that prices the company *high* ($V_p>\widehat{V}_p$) pushes the MFN SAFE into the seed round, where it converts at V and is borne by the seed investor. The pre-seed investor therefore chooses, through its own valuation, who pays for the accelerator's clause.

4.3 Pricing under the clause

The pre-seed investor's objective is unchanged in form, $\Pi_p(V_p)=\sigma_p\big(M(\pi_{\min}(V_p))/V_p-1\big)$, but $\pi_{\min}(\cdot)$ now depends on the regime. Since $\pi_{\min}=V$, Lemma 4 gives

$$\frac{\partial\pi_{\min}}{\partial V_p} = \begin{cases} -\dfrac{\pi_{\min}^2}{J+m}\cdot\dfrac{\sigma_p}{V_p^2} & \text{if } V_p>\widehat{V}_p, \\[2ex] -\dfrac{\pi_{\min}^2}{J}\cdot\dfrac{\sigma_p+m}{V_p^2} & \text{if } V_p<\widehat{V}_p, \end{cases}$$

and, proceeding exactly as in Proposition 2,

$$\Pi_p'(V_p) = \begin{cases} \dfrac{\sigma_p}{V_p^2}\left(\dfrac{\sigma_p f(\pi_{\min})\pi_{\min}^3}{(J+m)V_p}-M(\pi_{\min})\right) & \text{if } V_p>\widehat{V}_p, \\[2ex] \dfrac{\sigma_p}{V_p^2}\left(\dfrac{(\sigma_p+m) f(\pi_{\min})\pi_{\min}^3}{J\,V_p}-M(\pi_{\min})\right) & \text{if } V_p<\widehat{V}_p. \end{cases}$$

Substituting the cap-table identity of each regime—$(\sigma_p + m)/V_p = 1 - \rho_a^0 - J/\pi_{\min}$ in the low-price regime, $\sigma_p/V_p = 1 - \rho_a^0 - (J+m)/\pi_{\min}$ in the high-price regime—the two first-order conditions collapse onto a single family of equations, which is the observation the rest of the section rests on.

Definition 1. For $J > 0$ let

$$\Phi(\pi; J) \;:=\; (1-\rho_a^0)\,\pi^3 f(\pi) \;-\; J\Big(\pi^2 f(\pi) + M(\pi)\Big), \qquad \pi \in (0,1].$$

Lemma 5. *The interior optimum of the pre-seed investor satisfies*

$$\begin{aligned} \Phi(\pi_{\min}; J) = 0 &\quad \textit{in the low-price regime } V_p < \widehat{V}_p, \\ \Phi(\pi_{\min}; J+m) = 0 &\quad \textit{in the high-price regime } V_p > \widehat{V}_p. \end{aligned}$$

Lemma 5 is the pivot of the paper. The two regimes are governed by the *same* equation, evaluated at two different effective investment costs: J when the pre-seed investor absorbs the MFN SAFE into its own block, and $J + m$ when it pushes the MFN SAFE into the seed round. Everything that follows is a comparison of the two roots.

4.4 The two thresholds and their order

We first record a restriction without which no pre-seed round can occur at all.

Lemma 6. *If $J + m \geq 1 - \rho_a^0$, the pre-seed investor never participates.*

We therefore maintain $J + m < 1 - \rho_a^0$ throughout. No further restriction on f is needed: the log-concavity already assumed in (A1) delivers the threshold.

Lemma 7. *Fix J with $0 < J < 1 - \rho_a^0$. Then $\Phi(\cdot\,; J)$ has a unique root π_J in $\big(J/(1-\rho_a^0), 1\big)$, and $\Phi(\pi; J) < 0$ for $\pi < \pi_J$ and $\Phi(\pi; J) > 0$ for $\pi > \pi_J$ on that interval.*

Two roots are used from here on. Write $\pi_1 := \pi(J)$ for the root at the residual investment cost the pre-seed investor leaves and $\pi_2 := \pi(J+m)$ for the root at $J+m$; since $\pi(\cdot)$ is strictly increasing (Lemma 2), $\pi_1 < \pi_2$.

The two are the same problem at two investment costs, and that is the whole of the MFN channel. When the pre-seed investor absorbs the clause into its own block, the seed investor still has to fund J and the threshold solves $\Phi(\cdot\,; J) = 0$—the benchmark equation of Corollary 1 with ρ_a^0 in place of ρ_a, so the clause is threshold-neutral. When the claim is pushed into the round, the seed investor must clear $J + m$ out of the same capitalization table, and the threshold rises to π_2. Pushing the clause on therefore costs the firm the projects between π_1 and π_2. Which of the two prevails is the subject of Section 4.5.

Example 1 (uniform density)**.** Let $\pi \sim U[0,1]$, so $f \equiv 1$ and $M(x) = (1-x^2)/2$. Then

$$\Phi(\pi; J) = (1-\rho_a^0)\pi^3 - \tfrac{J}{2}\big(\pi^2+1\big),$$

and $f \equiv 1$ is log-concave. With $\rho_a^0 = 0.07$ and $J = 0.24$ one obtains $\pi_1 \approx 0.552$, and adding a tranche of $m = 0.025$ gives $\pi_2 \approx 0.574$: the gap is the set of projects financed when the pre-seed investor absorbs the clause and abandoned when it pushes it into the round. The pivot is $\bar{x} = \sqrt{2}-1 \approx 0.414$, so π_1 and π_2 both sit above it at $J = I$ and below it once the investor pre-funds heavily—which is why all three regimes appear in Figures 1–3. These are the parameters used there, and they are illustrative.

4.5 Who bears the clause: three regimes

Which conversion regime is played is decided by the amount, and two thresholds separate the three possibilities. Write

$$T_1 \;:=\; (1-\rho_a^0)\,\pi_1 - (J+m), \qquad T_2 \;:=\; (1-\rho_a^0)\,\pi_2 - (J+m), \tag{6}$$

so that $T_1 < T_2$ because $\pi_1 < \pi_2$. Each is the amount at which one of the two unconstrained optima leaves the region of prices that can support it: $\sigma_p \le T_1$ is $\pi_1 \ge \widehat{V}_p$, and $\sigma_p \le T_2$ is $\pi_2 \ge \widehat{V}_p$. Because $\sigma_p + J = b + I$ whatever the investor pre-funds, both comparisons are conditions on the residual need alone: $\sigma_p \le T_1$ if and only if $b + I + m \le (1-\rho_a^0)\pi_1$.

Proposition 9 (Pricing regimes)**.** *At a given amount, the pre-seed investor's optimal price, the threshold it engineers and the capitalization table it leaves are the following, with $k = 1-\rho_a^0$ throughout.*

(i) If $\sigma_p \le T_1$, the clause converts inside the pre-seed block:

$$V_p^* = \frac{\sigma_p + m}{k - J/\pi_1} < V, \qquad \pi^*_{\min} = \pi_1,$$

$$\rho_a^* = \rho_a^0 + \Big(k - \frac{J}{\pi_1}\Big)\frac{m}{\sigma_p+m}, \qquad \rho_p^* = \Big(k - \frac{J}{\pi_1}\Big)\frac{\sigma_p}{\sigma_p+m}, \qquad \rho_s^* = \frac{J}{\pi_1}.$$

(ii) If $T_1 < \sigma_p \le T_2$, the two prices coincide at $V_p^* = V = \widehat{V}_p = (J+m+\sigma_p)/k$, and

$$\pi^*_{\min} = \widehat{V}_p, \qquad \rho_a^* = \rho_a^0 + \frac{k\,m}{J+m+\sigma_p}, \qquad \rho_p^* = \frac{k\,\sigma_p}{J+m+\sigma_p}, \qquad \rho_s^* = \frac{k\,J}{J+m+\sigma_p}.$$

(iii) If $\sigma_p > T_2$, the clause converts into the round:

$$V_p^* = \frac{\sigma_p}{k - (J+m)/\pi_2} > V, \qquad \pi^*_{\min} = \pi_2,$$

$$\rho_a^* = \rho_a^0 + \frac{m}{\pi_2}, \qquad \rho_p^* = k - \frac{J+m}{\pi_2}, \qquad \rho_s^* = \frac{J}{\pi_2}.$$

In each case the round is priced at $V = \pi^*_{\min}$ and the conditional probability of a seed round is $1 - F(\pi^*_{\min})$.

The amount decides who pays for the clause. An investor advancing a small one absorbs it: the price it would name anyway is low enough that the accelerator's claim converts alongside its own, and the two share the block $k - J/\pi_1$ in the proportions $m : \sigma_p$. An investor advancing a large one prices high enough to push the claim into the round, where the seed investor carries it out of a table that must still fund J. Between the two it prices exactly at the round, and the three claims convert together on one base.

Read as instruments, the three regimes are three different contracts written on the same piece of paper. In (i) the accelerator's claim converts at a base strictly below the round, which is the payoff of a capped SAFE whose cap binds—the cap being the price the pre-seed investor names. In (ii) and (iii) it converts at the round itself, which is the payoff of an uncapped SAFE, and the clause is inert on the price margin. What separates those two is not the accelerator's instrument but what the pre-seed investor pays for its own block: the round price in (ii), more than it in (iii). The regimes are therefore ordered by the ratio V_p^*/V, below one in (i), equal to one throughout (ii), above one in (iii). The seed investor's stake is J/V in all three, so that ratio and the seed investor's stake move against each other.

What that does to financing follows from the ordering $\pi_1 < \pi_2$, and it is not the same in the three regimes.

Corollary 2 (The clause and the threshold)**.** *Compare each regime with the benchmark of Corollary 1 at the same residual investment cost J.*

(i) On $\sigma_p \leq T_1$ the clause is threshold-neutral: $\pi^*_{\min} = \pi(J)$, the seed investor's stake $J/\pi(J)$ and the financing probability are the benchmark's, and the clause moves m/V_p^* points of equity from the pre-seed investor to the accelerator and nothing else.

(ii) On $T_1 < \sigma_p \leq T_2$ it raises the threshold to $\widehat{V}_p \in (\pi_1, \pi_2]$, strictly above the benchmark, and $\widehat{V}_p$ rises one-for-one with the amount.

(iii) On $\sigma_p > T_2$ it raises the threshold to $\pi_2 = \pi(J + m)$: the benchmark threshold of a round that costs $J + m$ instead of J.

The probability of a seed round is therefore weakly lower under the clause, strictly so at every amount above T_1.

The three regimes are the whole of the MFN channel, and the clause is not one thing across them. It redistributes without cost in the first, and it destroys financing states in the other two, by a margin that grows with σ_p until it settles at the gap $\pi_2 - \pi_1$.

Which of them a firm meets is not a matter of taste, because the investor must also want to participate, and participation caps the amount. Where that cap falls relative to T_1 and T_2 is settled by one comparison: the position of the pivot $\bar{x}$, the solution of $M(\bar{x}) = \bar{x}$, against the two thresholds. Since $\pi(\cdot)$ is strictly increasing with $\pi(J^\dagger) = \bar{x}$, that is the position of $J^\dagger$ against J and $J + m$.

Proposition 10 (Participation). *The pre-seed investor participates if and only if $\sigma_p \le \bar{\sigma}_p^m$, where*

$$\bar{\sigma}_p^m = \begin{cases} W(J) - m & \text{if } J \ge J^\dagger, \text{ that is } \bar{x} \le \pi_1, \\ (1-\rho_a^0)\,\bar{x} - (J+m) & \text{if } J < J^\dagger \le J+m, \text{ that is } \pi_1 < \bar{x} \le \pi_2, \\ W(J+m) & \text{if } J+m < J^\dagger, \text{ that is } \pi_2 < \bar{x}, \end{cases}$$

and these caps satisfy $\bar{\sigma}_p^m \le T_1$, $T_1 < \bar{\sigma}_p^m \le T_2$ and $\bar{\sigma}_p^m > T_2$ respectively. A participating investor therefore reaches the first regime alone, the first two, or all three.

One comparison read at three heights, and it says something simple. Everything turns on whether the projects that survive the threshold are worth more than the price at which the marginal one is financed, $M(\pi)$ against π, and $\bar{x}$ is where the two cross. Below the pivot a claim is still worth engineering a higher threshold for; above it, raising the bar destroys more than it collects. An investor can therefore afford to price its way out of the clause only when its threshold sits below the pivot—which, by Proposition 3, is the pre-funding branch. One that merely covers the shock leaves $J = I \ge J^\dagger$ and never leaves the first regime; one that pre-funds drives J down, and with it π_1 and then π_2 below the pivot, until all three are open to it.

That is the MFN channel stated in full: threshold-neutral where the investor advances the shock and no more, threshold-raising where it pre-funds. The same cash that lowers the bar through J is what opens the regimes that raise it, and the two effects meet in Section 4.7.

4.6 When the pre-seed investor holds the same clause

Throughout, the pre-seed investor has held a *fixed-base* claim: it names a base V_p, buys $\rho_p = \sigma_p/V_p$, and keeps that share whatever the round price turns out to be. Suppose instead that it secures for itself the protection the accelerator holds against it—conversion at the more favourable of the two bases, so that its share becomes

$$\frac{\sigma_p}{\min(V, V_p)},$$

exactly the form of the accelerator's MFN SAFE. In market terms this is the standard capped SAFE, whose holder converts at the lower of its cap and the round price; we call it a *best-price* claim to keep the mechanism rather than the product name in view. The question is not which of two products is better, but what happens when the clause is held by an investor who could otherwise have set its own price.

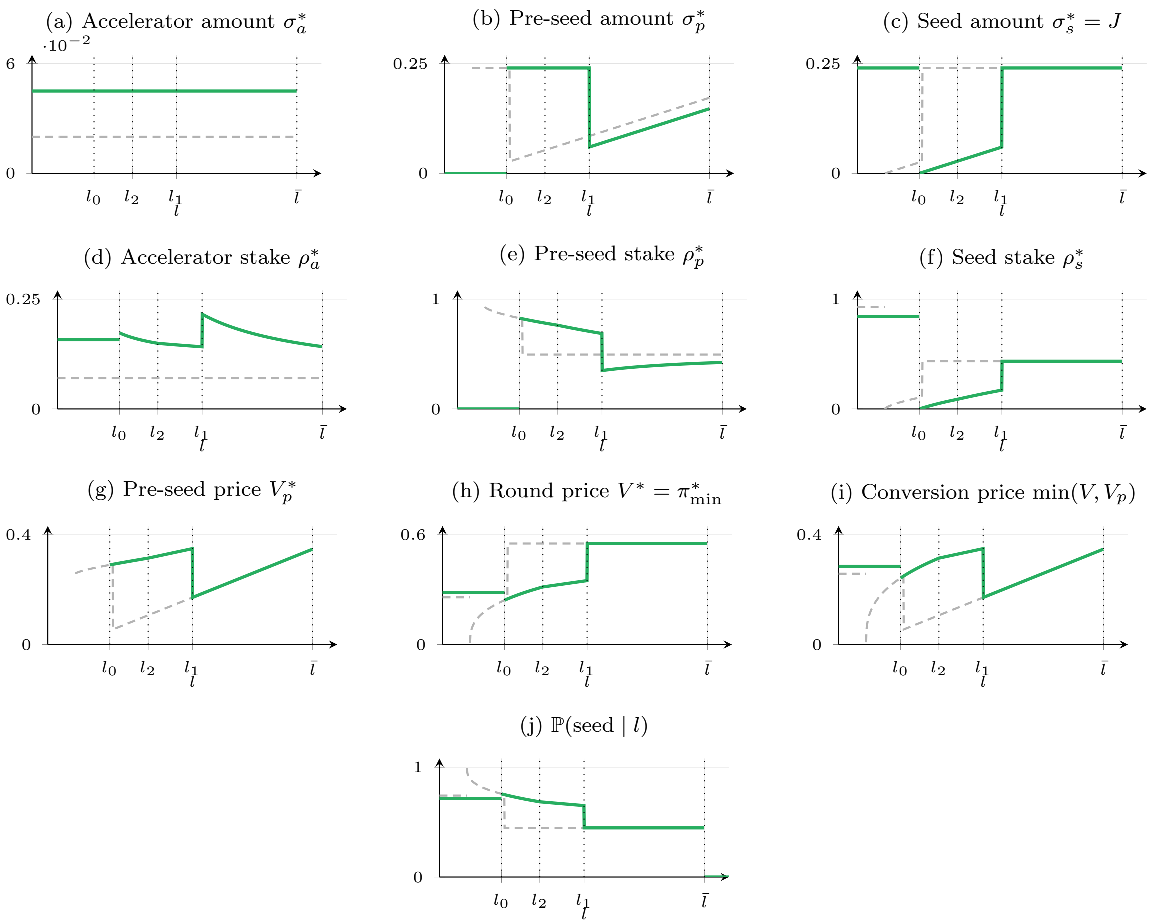

Figure 2: The same outcomes once the accelerator also holds an MFN SAFE, in green, against the benchmark of Figure 1 in dashed grey; $l_0 = \sigma_a + m$ is the entry point. The cash channel moves that point from σ_a to l_0. The MFN channel is panel (d): the accelerator's stake is no longer the constant ρ_a^0 but $\rho_a^0 + m/\min(V, V_p^*)$, worth most where the round it converts alongside is smallest. The three conversion regimes of Proposition 9 appear here in reverse order, and the reason is worth stating: the proposition orders them in the amount, but a budget-constrained investor that pre-funds advances the *same* amount β at every shock, so what moves with l is not σ_p but the cutoffs T_1, T_2, through the residual cost J. Small shocks therefore sit in regime (iii), where the SAFE is pushed into the round and the threshold in (h) is π_2; $[l_2, l_1]$ is regime (ii), where (g) and (h) coincide; and past l_1 the investor absorbs the clause in its own price and the threshold is π_1. Drawn for f uniform on $[0, 1]$, $\rho_a = \rho_a^0 = 0.07$, $I = 0.24$, $\sigma_a = 0.02$, $\beta = 0.24$, $m = 0.025$—the budget is the largest Assumption (B) allows, $\beta = I$, which is what puts all three conversion regimes within reach: a smaller one confines the investor to the first. The parameters are illustrative: the levels of the stakes reflect the monopolistic closure, and it is their shape in l that the figure is about.

The seed investor's problem. Both claims now convert at $\min(V, V_p)$, so the saturated capitalization table reads

$$\rho_a^0 + \frac{m}{\min(V, V_p)} + \frac{\sigma_p}{\min(V, V_p)} + \frac{J}{V} \;=\; 1,$$

and therefore

$$V \;=\; \begin{cases} \dfrac{J+m+\sigma_p}{1-\rho_a^0} \;=\; \widehat{V}_p & \text{if } V < V_p \quad \text{(the cap does not bind),} \\[2ex] \dfrac{J}{1-\rho_a^0 - \dfrac{m+\sigma_p}{V_p}} & \text{if } V > V_p \quad \text{(the cap binds),} \end{cases}$$

the first branch obtaining when $V_p \geq \widehat{V}_p$ and the second when $V_p < \widehat{V}_p$, exactly as in Lemma 4 and with the same cutoff. The asymmetry with the fixed-base case is immediate, and it is the whole of the argument: above $\widehat{V}_p$ the round price no longer depends on V_p. Once the cap stops binding, the pre-seed investor's pricing decision has no effect on anything.

The pre-seed investor's problem. The objective is $\Pi_p(V_p) = \sigma_p\big(M(\pi_{\min})/\min(V, V_p) - 1\big)$. Below $\widehat{V}_p$ the cap binds, conversion occurs at V_p, and $\pi_{\min} = J/(k-(m+\sigma_p)/V_p)$, so that

$$\frac{\partial \pi_{\min}}{\partial V_p} = -\frac{\pi_{\min}^2}{J} \cdot \frac{\sigma_p + m}{V_p^2} < 0, \qquad \Pi_p'(V_p) = \frac{\sigma_p}{V_p^2}\left(\frac{(\sigma_p+m)f(\pi_{\min})\pi_{\min}^3}{J\,V_p} - M(\pi_{\min})\right),$$

and substituting the capitalization identity $(m+\sigma_p)/V_p = k - J/\pi_{\min}$ returns $\Phi(\pi_{\min}; J) = 0$: the same equation as the low-price regime of Lemma 5, with the same interior optimum π_1. Above $\widehat{V}_p$ the cap does not bind, conversion occurs at $V = \widehat{V}_p$, and both price and threshold are constants,

$$\Pi_p(V_p) \;=\; \sigma_p\left(\frac{M(\widehat{V}_p)}{\widehat{V}_p} - 1\right) \qquad \text{for every } V_p \geq \widehat{V}_p,$$

so $\Pi_p' = 0$ on that entire range: every valuation above the cutoff is optimal and all of them pay the same.

Proposition 11 (Best-price regimes)**.** *Under a best-price claim the pre-seed investor's problem has two regimes rather than three:*

(i) if $\sigma_p \leq T_1$ the cap binds, the claim converts at $V_p^* = (\sigma_p+m)/(k-J/\pi_1)$, and the threshold is π_1, exactly as under a fixed-base claim;

(ii) if $\sigma_p > T_1$ the cap does not bind: every $V_p \geq \widehat{V}_p$ is optimal, all of them deliver the same payoff, and the threshold is $\widehat{V}_p$.

On small amounts the two instruments therefore coincide, and this is what makes the market's default coherent.

Remark 2 (A cap that never binds is a price)**.** On $\sigma_p \leq T_1$ the best-price claim prices strictly below the round, so the minimum is always its own base and it converts at V_p^*: a capped SAFE and a fixed-base SAFE are the *same contract* there. The cap is worth exactly the price its holder would have named, and choosing between the two instruments is immaterial.

The capped, no-discount SAFE is the dominant pre-seed structure, held by roughly three in four SAFE investors by late 2025 (Carta, 2025), and the model says that an investor advancing amounts below T_1 gives up nothing by holding it. By Proposition 10 that covers every amount available to an investor that merely bridges the shock. Past T_1 the two part company, and the comparison runs against the protection.

Proposition 12 (The clause constrains a holder who can price)**.** *The best-price claim is weakly dominated by the fixed-base claim for the pre-seed investor, and strictly dominated whenever $\sigma_p > T_2$. The domination is strict at an amount the investor would actually advance only if $\pi_1 < \bar{x}$; where the first regime is the only one participation allows, the two claims are worth the same at every such amount.*

The contrast with Proposition 9 is the point. Holding a fixed-base claim the investor has three regimes and still optimizes on large amounts: it prices high, pushes the accelerator's claim into the round, and attains π_2. Holding a best-price claim the high-price branch collapses—the cap hands the round the conversion, the threshold sticks at $\widehat{V}_p$, and the investor loses the ability to price at all.

The economics runs against instinct. The clause is a guarantee, and it delivers what it promises: its holder converts on the better of the two bases. But conversion is not a transfer from someone outside the company—it is a claim on the same capitalization table that must also accommodate J. By loading that table precisely in the states where the round prices low, the clause raises the threshold above which a round happens at all, and so destroys the states in which the claim pays anything. The protection works through a cap-table externality that turns against its own holder, and waiving it is a commitment device: it leaves room for the seed investor and buys back the financing states.

That is a statement about holders rather than about clauses. The accelerator commits at admission, before anything is known, and cannot revise its terms; for it the clause is pure gain, since it recovers a base it could never have negotiated. The pre-seed investor arrives after the shock and *sets* a price; for it the same clause substitutes a mechanical conversion rule for a decision it was making optimally, and the substitution can only weakly help. The condition $\pi_1 < \bar{x}$ says when the difference is more than weak: it is the pre-funding branch of Proposition 3, where the investor advances large amounts—and large amounts are exactly where a rule that fixes the conversion price costs the most.

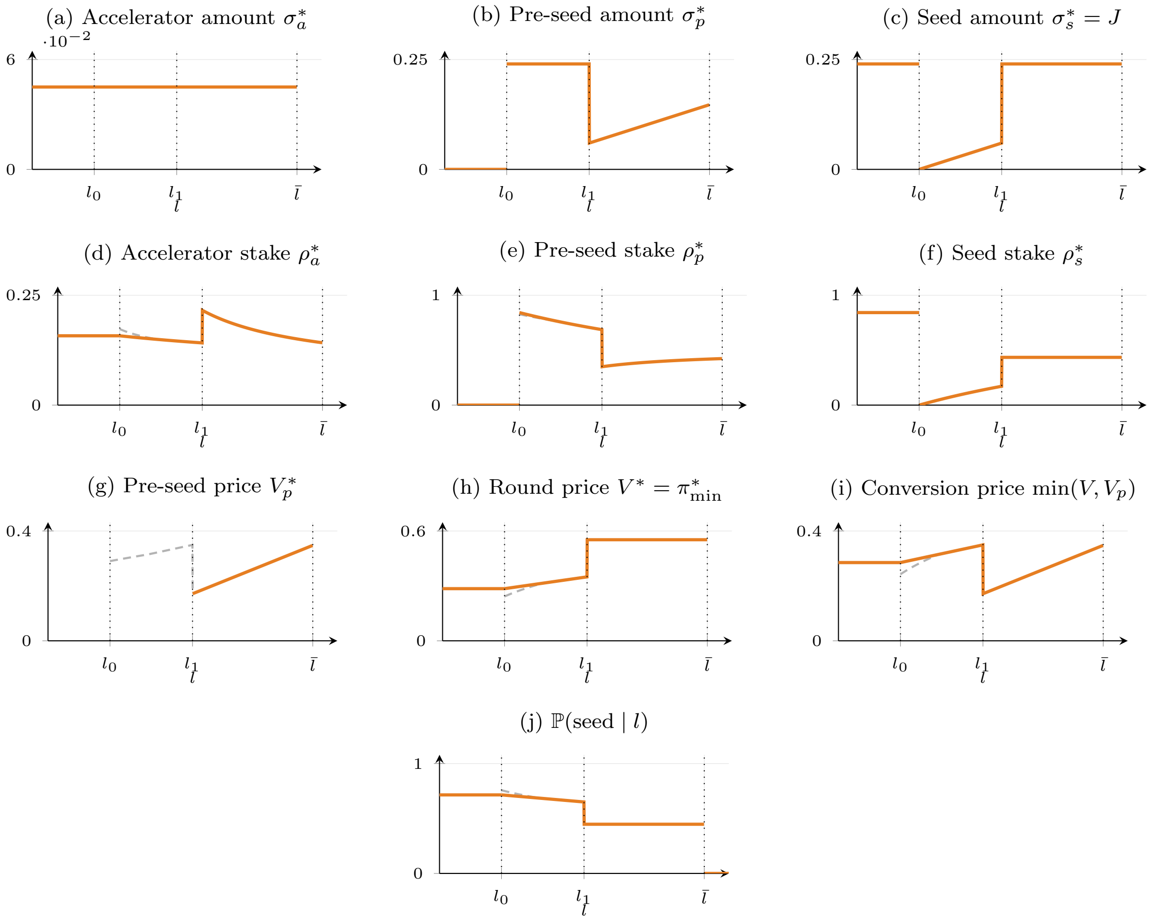

Figure 3: The same outcomes when the pre-seed investor holds a best-price claim rather than a fixed base, in orange, against the fixed-base case of Figure 2 in dashed grey. The two coincide wherever the cap binds, which by Proposition 11 is every shock past l_1: there the clause costs its holder nothing, which is Remark 2. Below l_1 the difference is Proposition 12: the fixed-base investor has a high-price branch and rides it at the threshold π_2, whereas the best-price investor has none—its claim converts at the round price whatever base it names, so (g) is undefined, the threshold in (h) is pushed up to $\widehat{V}_p$ and the seed probability in (j) is strictly lower. The protection binds exactly where it hurts. Drawn for f uniform on $[0, 1]$, $\rho_a = \rho_a^0 = 0.07$, $I = 0.24$, $\sigma_a = 0.02$, $\beta = 0.24$, $m = 0.025$—the budget is the largest Assumption (B) allows, $\beta = I$, which is what puts all three conversion regimes within reach: a smaller one confines the investor to the first. The parameters are illustrative: the levels of the stakes reflect the monopolistic closure, and it is their shape in l that the figure is about.

4.7 What the clause does to the amount

Propositions 9 and 10 hold the amount fixed. Once it is a choice again, the clause acquires a second margin, and it is not the one threshold neutrality would suggest.

Proposition 13. *In the first regime of Proposition 9, at amount σ_p and residual investment cost J,*

$$\frac{\partial \Pi_p}{\partial \Delta} \;=\; \frac{m\,W(J)}{(\sigma_p+m)^2} \;+\; \frac{\sigma_p}{\sigma_p+m}\,\Lambda(J) \;-\; 1,$$

which exceeds its no-clause counterpart $\Lambda(J)-1$ of Lemma 3 by

$$\frac{m\,M(\pi_1)}{\sigma_p+m}\left[\frac{1}{V_p^*}-\frac{1}{\pi_1}\right] \;\geq\; 0,$$

strictly whenever $V_p^* < V$. The MFN SAFE therefore weakly raises the pre-seed investor's incentive to pre-fund the seed round, strictly wherever it prices below the round.

The mechanism is a dilution asymmetry, and it is visible in the fraction $\sigma_p/(\sigma_p+m)$ alone. The MFN SAFE removes a *fixed number of units* from the pre-seed investor's block, so it costs a small amount proportionally far more than a large one, and advancing a bigger amount is among other things a way of diluting the clause. The clause, by taxing small amounts, pushes the investor toward exactly the behaviour that lowers the financing threshold, and this is the one margin on which the reform is not neutral.

The prediction about *amounts* is therefore unambiguous where the threshold predictions were not: pre-seed rounds should get larger, and larger by more than the mechanical substitution the cash channel produces. But the same force has a second destination, and it is the one the rest of the paper has been building toward.

Corollary 3 (The clause widens its own pre-funding region)**.** *The pivot of Proposition 3 falls under the clause, $\tilde{\Delta}^m \leq \tilde{\Delta}$, strictly wherever the investor prices below the round. The set of budgets at which the investor pre-funds rather than merely covering the shock is therefore larger with the clause than without it.*

Put the corollary beside Proposition 10 and the shape of the result appears. Pre-funding is what drives J down; J falling below $J^\dagger$ is what opens regimes (ii) and (iii); and those are the regimes in which the clause stops being threshold-neutral and starts destroying financing states. The clause taxes the small amount, the investor answers with a larger one, and a large enough amount is what carries the clause into the round. *The instrument pushes the investor toward the region in which the instrument ceases to be harmless.*

Two things keep this from being a clean welfare verdict, and both should be stated. The same pre-funding lowers J, and a lower J lowers the financing threshold within any given regime—so the identical move helps on one margin while opening the door on the other. And the door only opens if the budget reaches that far: an investor bounded well below I writes larger amounts under the clause and still never approaches T_1. The model therefore does not sign the sum. What it does is identify the quantity that decides it—how much intermediate investors pre-fund, relative to the round they bridge to—and say why that ratio, rather than the drafting of any clause, is what a regulator or an accelerator would have to look at.

Table 3 collects the section, putting the benchmark beside each of the three regimes. The accelerator's row is the novelty throughout: its stake is no longer the constant ρ_a but a share the pre-seed investor's own price determines. Below that row the columns part company. In regime (i) everything matches the benchmark—the seed investor advances the same amount, holds the same share, and finances the same set of projects. In (ii) and (iii) it does not: the seed stake and the round price move with the clause, and the set of financed projects shrinks. The amount itself is the corner of Proposition 3 evaluated at the accelerator's reformed contribution $\sigma_a + m$: the pre-seed investor advances the residual need $l - \sigma_a - m$ when its budget binds and β when it does not, so that $J = I$ in the first case and $J = I - \beta + l - \sigma_a - m$ in the second.

Read down the columns rather than across the rows: two prices generate every entry. Write V^* for the price of the round and $P^* := \min(V^*, V_p^*)$ for the price at which the accelerator's MFN SAFE converts—the better of the two, which is what the clause entitles it to. Then in all three regimes

$$\rho_a^* = \rho_a^0 + \frac{m}{P^*}, \qquad \rho_s^* = \frac{J}{V^*}, \qquad \rho_p^* = k - \frac{J}{V^*} - \frac{m}{P^*}, \qquad V_p^* = \frac{\sigma_p}{\rho_p^*}, \tag{7}$$

and the benchmark column is the same four expressions at $m = 0$. The three stakes sum to k by construction, so the pre-seed investor holds whatever the seed investor and the MFN SAFE leave, and the whole table moves with V^* and P^* alone. Both are pinned by the amount, and both rise with it: the MFN SAFE converts at the pre-seed price while that price sits below the round, at the common price on the middle range, and at the round beyond T_2.

Proposition 14 (The order of the table)**.** *Fix J and m and let the amount vary. Under (A1):*

(i) $\sigma_p \mapsto V^*$ and $\sigma_p \mapsto P^*$ are continuous and non-decreasing, strictly increasing on $(T_1, T_2]$ and on $(0, T_2]$ respectively. Every entry of (7) is therefore continuous at T_1 and at T_2, the two prices meeting at π_1 and at π_2 there.

(ii) Across the three regimes the accelerator's stake and the seed investor's stake fall, while the pre-seed stake, the pre-seed price, the round price and the threshold rise; the probability of a seed round falls with the last of these.

	Benchmark	With the MFN SAFE		
		(i) $\sigma_p \le T_1$	(ii) $T_1 < \sigma_p \le T_2$	(iii) $\sigma_p > T_2$
Accelerator σ_a^*	σ_a		$\sigma_a + m$	
Accelerator ρ_a^*	ρ_a	$\rho_a^0 + \Big(k - \dfrac{J}{\pi_1}\Big)\dfrac{m}{\sigma_p + m}$	$\rho_a^0 + \dfrac{k\,m}{J + m + \sigma_p}$	$\rho_a^0 + \dfrac{m}{\pi_2}$
Pre-seed σ_p^*		$l - \sigma_a$ (resp. $l - \sigma_a - m$) if the budget binds, β if not		
Pre-seed ρ_p^*	$k - \dfrac{J}{\pi_1}$	$\Big(k - \dfrac{J}{\pi_1}\Big)\dfrac{\sigma_p}{\sigma_p + m}$	$\dfrac{k\,\sigma_p}{J + m + \sigma_p}$	$k - \dfrac{J + m}{\pi_2}$
Seed $\sigma_s^* = J$		I if the budget binds, $I - \beta + l - \sigma_a$ (resp. $-m$) if not		
Seed ρ_s^*	$\dfrac{J}{\pi_1}$	$\dfrac{J}{\pi_1}$	$\dfrac{k\,J}{J + m + \sigma_p}$	$\dfrac{J}{\pi_2}$
Pre-seed price V_p^*	$\dfrac{\sigma_p}{k - J/\pi_1}$	$\dfrac{\sigma_p + m}{k - J/\pi_1}$	$\widehat{V}_p$	$\dfrac{\sigma_p}{k - (J + m)/\pi_2}$
Round price $V^* = \pi^*_{\min}$	π_1	π_1	$\widehat{V}_p$	π_2
$\mathbb{P}(\text{seed} \mid \cdot)$	$1 - F(\pi_1)$	$1 - F(\pi_1)$	$1 - F(\widehat{V}_p)$	$1 - F(\pi_2)$
Reached when	—	always	$J < J^\dagger$	$J + m < J^\dagger$

Table 3: Equilibrium with and without the accelerator's MFN SAFE, on the template of Table 2. Here $k = 1 - \rho_a$ in the benchmark and $k = 1 - \rho_a^0$ under the clause; $\pi_1 = \pi(J)$ and $\pi_2 = \pi(J + m)$ are the threshold map of Corollary 1 at the seed investor's residual cost and at that cost plus the tranche; and $\widehat{V}_p = (J + m + \sigma_p)/k$. The three columns are the regimes of Proposition 9, and the last row records which amounts a participating investor can advance (Proposition 10). Column (i) is the benchmark with one change: the total convertible share $k - J/\pi_1$ is the same object in both, split between the accelerator and the pre-seed investor in the proportions $m : \sigma_p$, so the seed stake, the round price and the financing probability are untouched. Columns (ii) and (iii) are not: the threshold rises, first with the amount and then to $\pi(J + m)$, and the financing probability falls with it. The accelerator's extra cash shifts the shock at which a pre-seed round becomes necessary, which is why the amount and the residual cost carry the m correction under the clause. Proposition 14 orders the columns of every row.

(iii) *Against the benchmark at the same amount, the same residual cost and an unchanged headline share $\rho_a^0 = \rho_a$: the pre-seed stake is strictly lower and the pre-seed valuation strictly higher in all three regimes, while the seed stake, the round price and the financing probability are the benchmark's in regime (i) and, in (ii) and (iii), leave the seed investor with a smaller stake bought at a higher threshold.*

Part (iii) holds J fixed and is therefore the price channel alone; the cash channel of Proposition 8 and the pre-funding channel of Proposition 13 move J itself, and the amount with it.

Two readings follow. The first is that the pre-seed investor is the only party that loses in every regime. Its stake is below the benchmark's throughout, and the mirror image of that loss is the row above it: the valuation at which it invests is above the benchmark's throughout, by the factor $\rho_p^{\text{bench}}/\rho_p^*$, and by most where the amount is smallest. Under the clause, that valuation is the price of handing m to the accelerator out of the same block.

The second reading is the one the regime labels conceal. Regime (i)—the regime in which the threshold is still $\pi(J)$, the seed investor's stake still $J/\pi(J)$ and the same projects still financed—is the regime in which the pre-seed stake is at its *lowest*. What regime (i) holds fixed is the threshold and the projects behind it, and it moves the capitalization table underneath them: the round survives the clause untouched precisely because the pre-seed investor absorbs it whole. Moving up through the regimes shifts part of that burden onto the seed investor, which pays it as a higher threshold; the pre-seed investor recovers stake in the same order, and the projects between π_1 and π_2 are what the transfer costs. The accelerator's own row is ordered against that: its stake is largest where the MFN SAFE converts inside the pre-seed block, at a price below the round, and smallest where it converts at the round. At a given residual cost, the instrument is worth most against the smallest amount.

5 Discussion

5.1 Empirical implications

The model delivers five predictions that do not require knowing the primitives. Table 4 collects them.

Valuations move with amounts, stakes do not. By Corollary 1, the pre-seed investor's equilibrium share is pinned down by (ρ_a, I, F) alone. Across companies from the same cohort facing different liquidity shocks, the amount raised at the pre-seed stage and the valuation at which it is raised should move together one-for-one, while pre-seed ownership stakes should be flat in the amount. This is a within-cohort prediction and therefore does not require comparing accelerators to one another. It is also the prediction that identifies the regime: it holds in the exact-funding corner and fails once the investor pre-funds, since the pre-funded part moves the threshold and hence the share.

The amount reveals the corner. By Proposition 3, an investor that merely covers a shock advances an amount that is small relative to the round that follows it, while an investor that pre-funds advances one of the order of the round itself. The ratio of the interim amount to the subsequent priced round is therefore a direct proxy for Δ/I, and reading it is the cheapest available test of which corner early-stage financing sits in—without estimating a single primitive. The same ratio predicts the cross-sectional pattern of thresholds: flat in the need where amounts are small, rising in the need where they are budget-constrained.

Pre-seed rounds get larger. By Proposition 13, the MFN SAFE removes a fixed sum from the pre-seed investor's block and therefore dilutes small amounts proportionally more than large ones, which strengthens the incentive to pre-fund. Post-2022 pre-seed rounds should thus be larger, and larger by more than the mechanical substitution the cash channel produces—the accelerator's extra cash reduces the residual need, which on its own would make interim amounts *smaller*. This is the one prediction of the reform whose sign the model does not leave ambiguous, and the two forces are separable in data: the cash channel shifts the need, the clause shifts the amount conditional on the need.

Seed success rises only where the clause is neutral. Two channels of the reform push seed success up and one pushes it down. The cash channel expands the region in which no pre-seed round is needed at all (Proposition 8), and the pre-funding channel of Proposition 13 lowers the seed investor's residual investment cost to the extent that the larger amounts it induces are pre-funding rather than padding. The price channel cuts the other way, and only past T_1: it is silent wherever the investor absorbs the clause (Corollary 2(i)) and adverse wherever the clause reaches the round, so the net effect is signed only for investors confined to the first regime—those leaving $J \geq J^{\dagger}$, by Proposition 10. The magnitude is an empirical question, and the natural design compares seed-graduation rates of comparable cohorts around January 2022, using companies from accelerators that did not change their terms as a control group. For cohorts whose investors merely bridge the shock, a null result on graduation with a positive result on the amount advanced is exactly what the model predicts; a negative one would point at investors pre-funding past $J^{\dagger}$.

The price ratio sorts the regimes. The predictions above are silent on which regime a given company sits in, and Proposition 9 says that one observable settles it: the pre-seed valuation divided by the valuation of the round that follows. Below one, the clause converts inside the pre-seed block and is threshold-neutral; at one, the three claims convert on a single base; above one, the clause has been pushed into the round. Because $\pi^*_{\min}$ rises along that order, the ratio rises exactly where the seed investor's stake J/V and the probability of a seed round fall, and none of the three quantities requires knowing J or the primitives. This is what turns the branch of the previous prediction from a latent type into a measured one: under the reform, seed-graduation rates should be flat among companies whose pre-seed round priced below the seed round and lower among those priced at or above it. What it demands of data is a price per share at both rounds for the same company, which rules out sources recording amounts alone.

5.2 What the clause achieves, and where

What the clause is designed to do and what it does to financing are two different things, and the model separates them cleanly.

The distributive objective is met in every regime. Whatever price the pre-seed investor sets, the MFN SAFE converts on the more favourable of the two bases, so the accelerator is never left holding worse terms than the investor that came after it. That is what the clause was written for, and it works.

The financing effect splits at $J^\dagger$. On the constrained branch, where the investor advances the shock and no more, no participating investor reaches T_1 (Proposition 10): the investor absorbs the clause in its own price, raising its valuation by exactly the MFN SAFE, and the financing threshold is the one that would have prevailed without the clause. The accelerator gains equity, the pre-seed investor gives up precisely that equity, and no project that would otherwise have been financed fails to be. On the pre-funding branch, amounts past T_1 become available and the threshold rises—to $\widehat{V}_p$ on the middle range, to $\pi(J+m)$ beyond T_2—so financing states are destroyed.

What decides is the size of the amount, not the drafting of the clause. The same instrument is neutral in the hands of an investor bridging a shock and costly in the hands of one pre-funding a round. Incidence is therefore a property of balance sheets, and a regulator or an accelerator revising the clause changes nothing about which of the two applies.

5.3 The three arguments for the reform

Three arguments are made for clauses of this kind. The analysis supports one with a qualification, supports the second for a different reason than the one given, and cannot speak to the third.

That the reform reduces fundraising pressure: supported, on one branch. The cash

Prediction	Source	What to measure	What the model says
Amounts move, stakes do not	Cor. 1	Pre-seed stake against the amount raised, within an accelerator cohort	Stake flat in the amount, valuation moving one-for-one with it; the pattern breaks once the investor pre-funds, which is what makes it identify the regime
The amount reveals the corner	Prop. 3	Interim amount over the next priced round, a proxy for Δ/I	Small ratio: exact funding, thresholds flat in the need. Ratio of the order of the round: pre-funding, thresholds rising in it
Pre-seed rounds get larger	Prop. 13	Amount size before and after January 2022	Up, and by more than the cash channel alone implies—that channel cuts the residual need and so pushes amounts the other way
Seed success, sign by branch	Props. 8, 2, 13	Seed-graduation rates of comparable cohorts around January 2022	Up or flat where the investor merely covers the shock—the cash and pre-funding channels raise it and the price channel is neutral; down where it pre-funds past $J^\dagger$ and the clause reaches the round
The price ratio sorts the regimes	Prop. 9	Pre-seed valuation over the valuation of the next round, company by company	Below one: clause absorbed, threshold neutral. At one: one conversion base. Above one: clause in the round, seed stake and graduation both lower

Table 4: The five predictions of Section 5, their source in the model, and what each requires from data. None needs the primitives (F, G, m) to be known: every entry is a within-cohort comparison, a cross-sectional shape, or a before-and-after on the January 2022 change of terms. The first two identify which corner of Proposition 3 early-stage financing sits in; the next two are the reform's effects, on amounts and on seed success; the last reads off which pricing regime a company is in, and so tells the fourth which companies to compare.

channel mechanically enlarges the set of firms that never need an interim round. Conditional on needing one, an investor in the first regime raises the valuation by exactly the MFN SAFE while its own stake falls to $\rho_p^* \sigma_p/(\sigma_p + m)$—less dilution at a higher price, which is what the argument asserted, and the repricing it implicitly assumes is what Corollary 2(i) delivers. Past T_1 the argument fails: the investor prices at or above the round and the firm is diluted more, not less.

That it raises seed success: supported, but not for the reason given. The argument credits the price guarantee. The price channel in fact contributes nothing where the clause is neutral and works against seed success where it is not. What raises seed success is the cash the accelerator commits and the larger interim amounts the clause induces (Propositions 8 and 13)—and those larger amounts are also what can carry an investor past $J^\dagger$. The reform helps on a margin it did not advertise, and the same margin can turn against it.

That it attracts more founders: outside the model. The environment closes with the seed investor extracting the residual, so the entrepreneur's ex-ante value is zero before and after the reform and the margin along which applicants would be drawn does not exist here. Evaluating that claim requires a closure with positive founder rents—a competitive seed stage, for instance—under which the founder's ex-ante value would inherit the same regime structure.

5.4 Limitations

Six modelling choices bound the analysis.

The pre-seed investor is a monopolist. Under competition its profit is driven to zero and its valuation is pinned at the participation frontier $M(\pi_{\min}) = V_p$ rather than by the first-order condition. The three branches of the price survive that change, since they come from the capitalization table and not from the objective; which branch a competitive investor lands on need not be the one a monopolist picks, and we have not solved for it. What we can say is that competition does not dissolve the pre-funding motive: with zero profit the terms sit on the participation frontier, but the firm still prefers the offer carrying the lowest threshold, which is the most pre-funded one, so pre-funding survives and its surplus accrues to the founder instead.

The seed investor extracts the residual equity. That is what closes the model at the last stage, and it is also what makes the founder margin unavailable—the entrepreneur's ex-ante value is zero whatever the contract upstream, which is why the third argument for the reform cannot be evaluated here.

The entrepreneur exerts no effort, at any stage. The probability of success π is drawn and revealed, not produced. Shares therefore carry no incentive role, and dilution costs the firm only through the financing constraint it tightens, never through the effort it discourages—which is what lets the capitalization table be read as pure accounting, and what sets the model apart from the security-design tradition that runs from Neher (1999) to Repullo and Suarez (2004), where the split of returns is the object precisely because it moves effort. The omission is tied to the previous one: with the seed investor extracting the residual, the entrepreneur holds nothing in equilibrium, so an effort decision would have nothing to respond to. Restoring effort therefore means restoring a founder stake, and the two extensions have to be made together. What would come of it is not obvious in either direction, and the regimes divide again: in regime (i) the clause moves equity between the two investors and leaves the rest of the table untouched, so an effort margin would register nothing; in (ii) and (iii) the seed stake moves, and it would.

Effort at more than one date would also carry the dilution backwards. Write $\pi = \pi(e_1, e_2)$ for effort supplied before and after the shock, and suppose the two are complements, $\pi_{12} > 0$—an entrepreneur who has already worked finds the next round of work cheaper. The interim round dilutes between the two dates, so it lowers e_2; by complementarity that lowers the return to e_1, and e_1 falls *in anticipation*, before any dilution has occurred. The cost of an interim round would then exceed what the static calculation shows and part of it would be borne upstream, which is a channel this model cannot express at all. A second loop points the same way: a higher threshold makes financing less likely, which lowers the return to effort, which lowers π and makes the threshold bind harder still. Both are amplification rather than addition, and both would weigh on regimes (ii) and (iii) alone.

No investor exerts effort either. Accelerators sell training, mentorship and investor networking alongside their capital (Zeisberger et al., 2017; Gonzalez-Uribe and Leatherbee, 2018), and pre-seed investors arriving after the shock know the company and can act on it; neither activity appears here, where investors supply cash and nothing else. This omission does not run into the closure the way the entrepreneur's does—investors hold strictly positive stakes in all three regimes—and it bears directly on the result. The clause moves equity from the pre-seed investor to the accelerator, that is from the party still in contact with the company to one whose program has ended. If equity is what motivates support, the transfer runs the wrong way, and it does so precisely in regime (i), where the model finds the clause purely redistributive. Whether that channel is large enough to overturn threshold neutrality is an open question, and answering it requires a model in which what investors do, and not only what they pay, enters the project's prospects.

The liquidity shock is exogenous. A firm anticipating the clause might adjust its burn rate, which would make l a choice variable and the regimes endogenous to the contract itself.

The budget β is a parameter rather than a choice, and the bound $\beta \leq I$ that defines the investor as intermediate is imposed rather than derived. Endogenizing it turns that investor into a fund choosing its size before observing deal flow, and the natural discipline is that a fund large enough to pre-empt every round is a seed fund and prices its capital accordingly—which is to say that the boundary between investor tiers, taken as given here, is itself an equilibrium object. That is the extension we would rank first, and for a reason the analysis makes precise: the budget decides how far the investor can pre-fund, hence which side of $J^\dagger$ the residual cost falls on, hence whether the clause is neutral or reaches the round. Deriving it rather than assuming it would say which investors the clause helps and which it constrains, instead of taking their identities as data.

Each of these extensions preserves the architecture—a threshold, a participation cap, and the three-regime structure of the clause—while changing who captures the surplus.

6 Conclusion

A most-favored-nation clause in an accelerator's contract is written to protect the accelerator against terms granted to later investors, and it does so mechanically. Its effect on financing, however, runs through an agent it does not bind: the pre-seed investor, who decides by its own pricing whether the clause converts inside its block or inside the priced round. That investor holds two margins, an amount and a price, and they are not independent. Cash advanced beyond the interim shock survives to the round and lightens what the seed investor must contribute, so pre-funding buys a lower financing threshold at a marginal value that *rises* as the round is pre-funded. The amount is therefore a corner, and which corner is a conclusion rather than an assumption.

The price margin resolves into three regimes, separated by two cutoffs in the amount. Below the first, the investor prices low and absorbs the clause: it raises its valuation by exactly the amount the clause converts, hands that equity to the accelerator, and leaves the financing threshold where the benchmark put it—the clause redistributes and does not ration. Above the second, it prices high and passes the clause to the seed investor, who must clear $J+m$ out of the same capitalization table; the threshold rises from $\pi(J)$ to $\pi(J+m)$ and financing states are destroyed. Between them the two prices bunch and the clause converts at the round. The contrast between the three is one of rationing; the redistribution is common to all of them, the pre-seed stake sitting strictly below its benchmark value at a given amount and the valuation strictly above. A higher interim valuation is what the clause produces in every state of the world.

Which regimes an investor can reach is settled by one comparison, and this is the paper's sharpest statement. Participation caps the amount, and where that cap falls depends on whether the cost the investor leaves the seed round lies above or below $J^\dagger$, the point at which pre-funding stops paying. An investor that merely bridges the shock is on the far side of the pivot and never leaves the first regime: for it the clause is threshold-neutral. One that pre-funds heavily crosses the pivot and opens the other two. The same cash that lowers the financing bar through J is what opens the regimes that raise it, so the clause is neutral exactly where pre-funding does not pay, and adverse exactly where it does. What decides the incidence is not the drafting of the clause but the size of the amount—and therefore the balance sheet behind it.

The clause also has an unambiguous effect on that amount. Because it removes a fixed sum from a block whose size the investor chooses, it dilutes a small amount far more than a large one, and pushes the investor toward advancing larger ones. Larger amounts are the pre-funding that lowers the threshold; they are also what carries an investor across the pivot. The reform's two effects therefore run in opposite directions along the same margin, and the model does not sign their sum.

Finally, the value of the protection depends on who holds it. The accelerator commits at admission, on terms it can no longer revise, and for it a clause carrying no base of its own is pure gain: it recovers a price it could never have negotiated. The pre-seed investor arrives after the shock and *sets* a price; for it the same rule replaces a decision it was taking optimally with a mechanical outcome, and it is weakly worse off—strictly, by Proposition 12, wherever its amounts are large enough for the substitution to bind. The rule protects an investor that cannot price and constrains one that can.

The general lesson is about sequence. Protection granted at one financing stage is never neutral for the stages that follow, because the investors at those stages adjust the terms they offer, and they adjust on every margin they hold. Whether a reform of this kind helps the companies it is meant to help therefore turns on an empirical magnitude the theory identifies but cannot supply: how much intermediate investors pre-fund. The model says where to look—the ratio of the interim amount to the round that follows it—and what the answer would mean.

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A Proofs

Proof of Proposition 1

Proof. The objective is strictly increasing in the share $\frac{J}{V}$, so the cap-table constraint binds; and the seed investor proposes if and only if its payoff is non-negative. $\square$

Proof of Lemma 1

Proof. Any $\sigma_p < b$ leaves the shock uncovered. The firm is then liquidated before π is revealed, the claim never converts, and the amount is sunk, for a payoff of $-\sigma_p$; this is strictly dominated by not proposing, which yields zero. $\square$

Proof of Lemma 2

Proof. Bounds. $D > 0$ because $M > 0$ on $(0, 1)$, and writing (5) as

$$\psi(\pi) \;=\; k\,\pi\,\frac{\pi^2 f(\pi)}{\pi^2 f(\pi) + M(\pi)} \;=\; \frac{k\,\pi}{1 + M(\pi)/\big(\pi^2 f(\pi)\big)} \tag{8}$$

gives $0 < \psi(\pi) < k\pi < k$ at once, since $M > 0$.

Monotonicity. Using $M' = -\pi f$, so that $D' = \pi f + \pi^2 f'$,

$$\frac{\psi'(\pi)\,D(\pi)^2}{k} \;=\; \pi^2\big(3f + \pi f'\big)D \;-\; \pi^3 f \cdot \pi\big(f + \pi f'\big) \;=\; \pi^2\Big[\,2\pi^2 f^2 \;+\; M\big(3f + \pi f'\big)\Big],$$

the terms in ff' cancelling, all functions evaluated at π. Write $c := -f'(\pi)/f(\pi)$, so the bracket equals $f\big(2\pi^2 f + M(3 - c\pi)\big)$ and the claim reduces to

$$2\pi^2 f(\pi) \;+\; M(\pi)\big(3 - c\,\pi\big) \;>\; 0.$$

If $c \le 3/\pi$ the second term is non-negative and the inequality is immediate, since $f > 0$. Suppose then $c > 3/\pi$, in particular $c > 0$. Log-concavity of f in (A1) means that $\log f$ lies below its tangent at π, so $f(t) \le f(\pi)e^{-c(t-\pi)}$ for every $t \ge \pi$, whence

$$M(\pi) = \int_\pi^1 t\,f(t)\,dt \;\le\; f(\pi)\int_\pi^\infty t\,e^{-c(t-\pi)}\,dt \;=\; f(\pi)\,\frac{c\pi + 1}{c^2}.$$

Because $3 - c\pi < 0$, substituting this upper bound can only lower the left-hand side:

$$2\pi^2 f + M\big(3 - c\pi\big) \;\ge\; \frac{f}{c^2}\Big(2c^2\pi^2 + (c\pi + 1)(3 - c\pi)\Big) \;=\; \frac{f}{c^2}\Big((c\pi + 1)^2 + 2\Big) \;>\; 0.$$

Hence $\psi' > 0$ on $(0, 1)$.

Limits. At the lower end, $0 < \psi(\pi) < k\pi$ gives $\psi(0^+) = 0$.

At the upper end, write $r := (1-F)/f$ for the reciprocal of the hazard rate, non-increasing by (A1) and non-negative, so it has a limit $L \geq 0$ at 1^-. Suppose $L > 0$. Then $r \geq L$ on $(0,1)$, so $-\big(\log(1-F)\big)' = 1/r \leq 1/L$ there, and integrating from any $\pi_0 < 1$ gives $1 - F(\pi) \geq \big(1 - F(\pi_0)\big)e^{-(\pi-\pi_0)/L}$ for every $\pi < 1$. The right-hand side is bounded away from zero as $\pi \to 1^-$, contradicting $F(1^-) = 1$. Hence $L = 0$. Now $t \leq 1$ throughout $[\pi, 1]$, so $M(\pi) = \int_\pi^1 t\, f(t)\, dt \leq \int_\pi^1 f(t)\, dt = 1 - F(\pi)$—the value carried by the projects above the threshold is at most their probability—and therefore

$$0 \;\leq\; \frac{M(\pi)}{\pi^2 f(\pi)} \;\leq\; \frac{r(\pi)}{\pi^2} \;\xrightarrow[\pi\to 1^-]{}\; 0,$$

and (8) gives $\psi(1^-) = k$.

The three claims. Being continuous and strictly increasing with limits 0 and k, ψ is a bijection from $(0,1)$ onto $(0,k)$. Fix J in that range. Then $J = \psi(\pi)$ has the unique solution $\pi(J) = \psi^{-1}(J)$, and evaluating $\psi(\pi) < k\pi$ at $\pi = \pi(J)$ reads $J < k\,\pi(J)$: this is (i). Next $\varphi_J' = -D(\psi - J)/\pi^2$ has the sign of $J - \psi$, so φ_J strictly increases below $\pi(J)$ and strictly decreases above it, making the critical point the maximum: this is (ii). Finally the inverse of a strictly increasing map is strictly increasing: this is (iii). □

Proof of Lemma 3

Proof. $W(J) = \max_\pi \varphi_J(\pi)$ with $\partial\varphi_J/\partial J = -M(\pi)/\pi$ from (2); the envelope theorem evaluates this derivative at the maximizer $\pi = \pi(J)$, giving $W'(J) = -\Lambda(J)$. Since $\sigma_p = b + \Delta$ and $J = I - \Delta$, (3) gives $\partial\Pi_p/\partial\Delta = -W'(J) - 1 = \Lambda(J) - 1$. For monotonicity, $\pi(\cdot)$ is strictly increasing by Lemma 2—and $x \mapsto M(x)/x$ is strictly decreasing, since M is positive and strictly decreasing. Hence Λ is strictly decreasing in J and strictly increasing in Δ. □

Proof of Proposition 3

Proof. Two objects. Since $\Lambda(J) = M(\pi(J))/\pi(J)$, the marginal value of pre-funding crosses one where the threshold reaches the fixed point of M. Define $\bar{x}$ by $M(\bar{x}) = \bar{x}$: as $M(x) - x$ is continuous and strictly decreasing from $M(0) = \mathbb{E}(\pi) > 0$ at $x = 0$ to -1 at $x = 1$, that pivot exists and is unique in $(0,1)$. Define $J^\dagger$ by $\pi(J^\dagger) = \bar{x}$: by Lemma 2, $\pi(\cdot)$ is a strictly increasing bijection from $(0,k)$ onto $(0,1)$, so $J^\dagger = \psi(\bar{x})$ exists and is unique as well. Write $\Delta^\dagger := I - J^\dagger$, so that $\Lambda(J^\dagger) = 1$ by construction.

If $\Delta^\dagger \leq 0$ (i.e. $I \leq J^\dagger$), then $\Pi_p'(0) = \Lambda(I) - 1 \geq 0$ and, by convexity (Lemma 3), $\Pi_p' \geq 0$ throughout $[0, I]$: Π_p is non-decreasing, so $\sigma_p^* = \beta$ for every admissible β and $\tilde{\sigma}_p = b$.

If $\Delta^\dagger \in (0, I)$, then Π_p strictly decreases on $[0, \Delta^\dagger]$ and strictly increases on $[\Delta^\dagger, I]$ by Lemma 3, so $\sigma_p^* = b$ whenever $\beta - b \le \Delta^\dagger$. On $(\Delta^\dagger, I]$, let $h(\Delta) := \Pi_p(\Delta) - \Pi_p(0)$; h is strictly increasing there with $h(\Delta^\dagger) < 0$. If $h(I) \le 0$, $\sigma_p^* = b$ throughout and $\tilde{\sigma}_p = b + I$; otherwise h has a unique zero $\tilde{\Delta} \in (\Delta^\dagger, I)$, and comparing $\Pi_p(\beta - b)$ to $\Pi_p(0)$—the two candidates on $[0, \beta - b]$, since a convex function on an interval is maximized at an endpoint—is comparing $h(\beta - b)$ to zero, giving $\sigma_p^* = b$ for $\beta \le b + \tilde{\Delta}$ and $\sigma_p^* = \beta$ for $\beta > b + \tilde{\Delta}$, so $\tilde{\sigma}_p = b + \tilde{\Delta}$.

In every case $\tilde{\Delta} := \tilde{\sigma}_p - b$ is pinned down by $\Pi_p(\cdot) - \Pi_p(0)$ alone, and by (3), $\Pi_p(\Delta) - \Pi_p(0) = W(I - \Delta) - b - \Delta - \big(W(I) - b\big) = W(I - \Delta) - W(I) - \Delta$, in which b cancels: $\tilde{\Delta}$ is the same number for every need b. □

The pre-emption counterfactual of Remark 1

Proof. Drop the bound of Assumption (B) and take $\beta \ge b + I$: the admissible range becomes $[0, I]$, the corner is $\Delta = I$ with $J = 0$, and the threshold equation $J = k\,\pi^3 f(\pi)/\big(M(\pi) + \pi^2 f(\pi)\big)$ forces $\pi \to 0$; in that limit $J/\pi = k\,\pi^2 f(\pi)/\big(M(\pi) + \pi^2 f(\pi)\big) \to 0$ because $M(0) = \mathbb{E}(\pi) > 0$, hence $\rho_p^* = k - J/\pi(J) \to k = 1 - \rho_a$ and $W(0) = (1 - \rho_a)\mathbb{E}(\pi)$. With $\sigma_p = b + I$ this gives the stated payoff. □

Proof of Proposition 2

Proof. By Proposition 1, $\pi_{\min} = J/(1 - \rho_a - \rho_p)$ with $\rho_p = \sigma_p / V_p$, so

$$\frac{\partial \pi_{\min}}{\partial V_p} \;=\; -\,\frac{\pi_{\min}^2}{J} \cdot \frac{\sigma_p}{V_p^2} \;<\; 0\,:$$

a more generous pre-seed valuation dilutes the entrepreneur less, leaves more room in the cap table for the seed investor, and therefore lowers the seed threshold. Differentiating Π_p and using $M'(x) = -x f(x)$,

$$\Pi_p'(V_p) \;=\; \sigma_p\left(\frac{M'(\pi_{\min})\,\partial\pi_{\min}/\partial V_p}{V_p} - \frac{M(\pi_{\min})}{V_p^2}\right) \;=\; \frac{\sigma_p}{V_p^2}\left(\frac{\sigma_p\, f(\pi_{\min})\,\pi_{\min}^3}{J\,V_p} - M(\pi_{\min})\right).$$

Setting $\Pi_p'(V_p) = 0$ yields the stated first-order condition. The derivation holds at a given amount, so Δ and hence J are fixed throughout. □

Proof of Corollary 1

Proof. Substitute $V_p = \sigma_p / \rho_p$ into Proposition 2: the factor σ_p cancels on both sides, leaving $f(\pi)\,\pi^3 \rho_p = J\,M(\pi)$. Replacing $\rho_p = 1 - \rho_a - J/\pi$ and rearranging gives the stated equation. The cancellation of σ_p is what the invariance statement rests on, and it is why the statement is conditional: J survives the cancellation, and J moves with the pre-funded part of the amount. $\square$

Proof of Proposition 4

Proof. Participation requires $\Pi_p(V_p^*) \geq 0$, i.e. $M(\pi_{\min}^*) \geq V_p^*$. Multiplying both sides by $\rho_p^* > 0$ and using $V_p^* \rho_p^* = \sigma_p$ gives the condition; ρ_p^* and $\pi_{\min}^*$ are the constants of Corollary 1. $\square$

Proof of Proposition 5

Proof. Take a constrained pre-seed investor, so that wherever a round occurs Proposition 3 gives $\sigma_p^* = b$, hence $\Delta = 0$ and $J = I$, and recall $b = l - \sigma_a$.

Region (N), $l \leq \sigma_a$. The accelerator's amount already absorbs the shock, so $b \leq 0$: there is no residual need to fund and no claim on which a pre-seed investor could be repaid, and $(\rho_p, \sigma_p) = (0, 0)$. The accelerator's amount is committed before the shock is known and is consumed by the interim stage, so nothing is carried into the round and $J = I$. Proposition 1 with $\rho_p = 0$ then prices the round at $V = I/(1 - \rho_a) = \pi_0$ and finances exactly the projects $\{\pi \geq \pi_0\}$, an event of probability $1 - F(\pi_0)$ conditional on this region.

Region (P), $\sigma_a < l \leq \sigma_a + \bar{\sigma}_p$. Here $0 < b \leq \bar{\sigma}_p$, so the participation condition of Proposition 4 holds and a round occurs; the investor advances at least b by Lemma 1 and exactly b by Proposition 3, so $J = I$. The price is the one of Proposition 2 and the threshold it engineers is $\pi_{\min}^* = \pi(I)$, which by Corollary 1 does not depend on b. The capitalization table binds (Proposition 1), so $\rho_p^* = 1 - \rho_a - I/\pi_{\min}^*$, the seed investor funds exactly $\{\pi \geq \pi_{\min}^*\}$, and the conditional probability is $1 - F(\pi_{\min}^*)$.

Region (L), $l > \sigma_a + \bar{\sigma}_p$. Now $b > \bar{\sigma}_p = W(I)$, so by (3) the payoff $W(I) - b$ is strictly negative. Since $W(I)$ is the maximum of $\Pi_p + \sigma_p$ over V_p, the bound binds at every price; and advancing less than b is dominated (Lemma 1). No round occurs, the shock is not absorbed, the firm is liquidated before π is revealed, and the conditional probability of a seed round is 0.

Ordering. Both thresholds are the residual investment cost divided by the equity the capitalization table leaves the seed investor, $\pi_0 = I/(1-\rho_a)$ and $\pi_{\min}^* = I/(1-\rho_a-\rho_p^*)$, so $\pi_0 < \pi_{\min}^*$ is exactly $\rho_p^* > 0$. That inequality is Lemma 7 applied with $k = 1 - \rho_a$: the root $\pi(I)$ lies strictly above $I/(1 - \rho_a)$, whence $\rho_p^* = 1 - \rho_a - I/\pi(I) > 0$. $\square$

Proof of Proposition 6

Proof. Condition on the three regimes of Proposition 5 and use

$$\mathbb{P}(\text{seed}) = \sum_R \mathbb{P}(R)\,\mathbb{P}(\text{seed} \mid R), \qquad \mathbb{P}(\text{seed} \mid \text{L}) = 0.$$

□

Proof of Proposition 7

Proof. Let $\beta > \tilde{\sigma}_p$. Wherever the investor participates, Proposition 3 gives $\sigma_p^* = \beta$, so $\Delta = \beta - b$ and $J(b) = I - \beta + b$, strictly increasing in b and, by (B), strictly positive on the relevant range.

The region. Two conditions are needed. First the shock must be coverable: the amount must reach b by Lemma 1 and cannot exceed β by (B), so $b \leq \beta$. Second the claim must be worth its amount: by (3) the maximized payoff at $\sigma_p = \beta$ is $\Pi_p = W\big(J(b)\big) - \beta$, so participation requires $W\big(J(b)\big) \geq \beta$. By Lemma 3, $W' = -\Lambda < 0$, and $J(\cdot)$ is strictly increasing, so $b \mapsto W(J(b))$ is strictly decreasing on $[0, \beta]$: the second condition is an upper bound on b. If $W(I) \geq \beta$ it never binds and the region is $b \leq \beta$; otherwise, provided $W(I - \beta) \geq \beta$, strict monotonicity gives a unique root $b^W \in [0, \beta)$ of $W(J(b^W)) = \beta$ and the condition reads $b \leq b^W$; if $W(I - \beta) < \beta$ no shock admits a round. Writing $b^W := \beta$ in the first case, the two conditions combine into $b \leq \bar{b} = \min\{\beta, b^W\}$, that is $\sigma_a < l \leq \sigma_a + \bar{b}$.

The threshold. On that region Corollary 1 applies at the residual investment cost $J(b)$ and delivers $\pi\big(J(b)\big)$, strictly increasing in b because $J(\cdot)$ is strictly increasing and $\pi(\cdot)$ is strictly increasing (Lemma 2). The threshold therefore varies across the region where a round occurs: each shock carries its own.

The probability. The region $l \leq \sigma_a$ is unchanged from Proposition 5: no round, $J = I$, threshold π_0, contributing $G(\sigma_a)\big(1 - F(\pi_0)\big)$. On $(\sigma_a, \sigma_a + \bar{b}\,]$ the conditional probability at shock l is $1 - F\big(\pi(J(l - \sigma_a))\big)$, and integrating against dG gives the second term; the shock and project distributions no longer separate, so the product of the constrained case becomes an integral. Beyond $\sigma_a + \bar{b}$ the firm is liquidated and the contribution is zero. □

Proof of Proposition 8

Proof. By Corollary 1, raising σ_a leaves π_0, $\pi^*_{\min}$, ρ_p^* and $\bar{\sigma}_p$ unchanged: they depend on (ρ_a, I, F) only. What changes is the partition of the shock space in Proposition 5: the no-pre-seed region $\{l \leq \sigma_a\}$ expands and the liquidation region $\{l > \sigma_a + \bar{\sigma}_p\}$

shrinks, both by the same displacement. Differentiating Proposition 6,

$$\frac{\partial\,\mathbb{P}(\text{seed})}{\partial\sigma_a} = g(\sigma_a)\big(F(\pi^*_{\min}) - F(\pi_0)\big) + g(\sigma_a+\bar{\sigma}_p)\big(1 - F(\pi^*_{\min})\big) \;>\; 0,$$

where the first term is non-negative because $\pi_0 < \pi^*_{\min}$ (Proposition 5) and the second is strictly positive. □

Proof of Lemma 4

Proof. Take the first branch. $V < V_p$ reads $(J+m)/\big(1-\rho_a^0-\sigma_p/V_p\big) < V_p$, and since the denominator is positive whenever a seed round occurs, this is equivalent to $J+m < (1-\rho_a^0)V_p - \sigma_p$, that is to $V_p > (J+m+\sigma_p)/(1-\rho_a^0)$. The second branch is the same computation with m moved into the pre-seed block: $V > V_p$ reads $J > (1-\rho_a^0)V_p - (\sigma_p+m)$, i.e. $V_p < \widehat{V}_p$. □

Proof of Lemma 5

Proof. In the low-price regime, set $\Pi_p' = 0$: $(\sigma_p+m)f(\pi)\pi^3 = JV_pM(\pi)$. Divide by V_p and use $(\sigma_p+m)/V_p = 1-\rho_a^0 - J/\pi$:

$$\Big(1-\rho_a^0-\tfrac{J}{\pi}\Big)f(\pi)\pi^3 = J\,M(\pi) \iff (1-\rho_a^0)\pi^3 f(\pi) - J\pi^2 f(\pi) - JM(\pi) = 0,$$

which is $\Phi(\pi;J)=0$. In the high-price regime the same steps with $\sigma_p/V_p = 1-\rho_a^0 - (J+m)/\pi$ and the factor $(J+m)$ in place of J give $\Phi(\pi;J+m)=0$. □

Proof of Lemma 6

Proof. Suppose a pre-seed round occurs. If $V_p > \widehat{V}_p$ then $\pi_{\min} = (J+m)/(1-\rho_a^0 - \sigma_p/V_p) \geq (J+m)/(1-\rho_a^0) \geq 1$, which is impossible since $\pi_{\min} \leq 1$. If $V_p \leq \widehat{V}_p$ then $\pi_{\min} = J/(1-\rho_a^0-(\sigma_p+m)/V_p)$ is decreasing in V_p, hence

$$\pi_{\min} \;\geq\; \pi_{\min}(\widehat{V}_p) = \frac{J}{1-\rho_a^0 - \dfrac{(\sigma_p+m)(1-\rho_a^0)}{J+m+\sigma_p}} = \frac{J+m+\sigma_p}{1-\rho_a^0} \;\geq\; \frac{J+m}{1-\rho_a^0} \;\geq\; 1,$$

again impossible. Hence $J+m < 1-\rho_a^0$ whenever a pre-seed round occurs. □

Proof of Lemma 7

Proof. Definition 1 factors as $\Phi(\pi;J) = D(\pi)\big(\psi(\pi)-J\big)$, with $D = \pi^2 f + M > 0$ and ψ the map of Lemma 2 read at $k = 1-\rho_a^0$. That lemma makes ψ a strictly increasing bijection from $(0,1)$ onto $(0,k)$, so for $0 < J < 1-\rho_a^0$ the equation $\psi(\pi) = J$ has exactly one solution π_J, which is therefore the unique root of $\Phi(\cdot\,; J)$; it lies in $\big(J/(1-\rho_a^0), 1\big)$ because $\psi(\pi) < k\pi$; and Φ, having the sign of $\psi - J$, is negative below it and positive above it. $\square$

Proof of Proposition 9

Proof. Write $k := 1-\rho_a^0$. The amount σ_p, hence Δ and $J = I - \Delta \in (0, I]$, are held fixed; $\pi_1 = \pi(J)$ and $\pi_2 = \pi(J+m)$ are the roots named after Lemma 7, both well defined under Lemma 6, with $\pi_1 < \pi_2$ because $\pi(\cdot)$ is strictly increasing (Lemma 2).

Step 1: choosing a price is choosing a threshold. The round is priced at the marginal project, $V = \pi_{\min}$ (Proposition 1), and the capitalization identity of each conversion regime makes $\pi_{\min}$ continuous and strictly decreasing in V_p:

$$\pi_{\min} \;=\; \frac{J}{k-(\sigma_p+m)/V_p} \quad \text{on } V_p \le \widehat{V}_p, \qquad \pi_{\min} \;=\; \frac{J+m}{k-\sigma_p/V_p} \quad \text{on } V_p \ge \widehat{V}_p.$$

The two agree at the cutoff of Lemma 4, where both equal $\widehat{V}_p$ itself: substituting $V_p = \widehat{V}_p = (J+m+\sigma_p)/k$ gives $k-(\sigma_p+m)/\widehat{V}_p = kJ/(J+m+\sigma_p)$ and $k-\sigma_p/\widehat{V}_p = k(J+m)/(J+m+\sigma_p)$, so each expression returns $(J+m+\sigma_p)/k$. The investor's problem can therefore be written over $\pi_{\min}$ instead of V_p, with the low-price regime $V_p \le \widehat{V}_p$ mapping onto $\pi_{\min} \ge \widehat{V}_p$ and the high-price regime onto $\pi_{\min} \le \widehat{V}_p$.

Step 2: the payoff on each regime. With $\varphi_J(\pi) = (k - J/\pi)M(\pi)$ from (2), read at $k = 1-\rho_a^0$, substitute the capitalization identity into $\Pi_p = \sigma_p\big(M(\pi_{\min})/V_p - 1\big)$. In the low-price regime $(\sigma_p+m)/V_p = k - J/\pi_{\min}$, so $\sigma_p/V_p = \frac{\sigma_p}{\sigma_p+m}\big(k - J/\pi_{\min}\big)$; in the high-price regime $\sigma_p/V_p = k - (J+m)/\pi_{\min}$ directly. Hence

$$\Pi_p \;=\; \begin{cases} \dfrac{\sigma_p}{\sigma_p+m}\,\varphi_J(\pi_{\min}) \;-\; \sigma_p & \text{on } \pi_{\min} \ge \widehat{V}_p \quad \text{(the clause converts at } V_p\text{)},\\[2ex] \varphi_{J+m}(\pi_{\min}) \;-\; \sigma_p & \text{on } \pi_{\min} \le \widehat{V}_p \quad \text{(the clause converts at } V\text{)}.\end{cases}$$

The two branches agree at $\pi_{\min} = \widehat{V}_p$, where both equal $\sigma_p\big(M(\widehat{V}_p)/\widehat{V}_p - 1\big)$, by the same substitution as in Step 1. Since $\sigma_p/(\sigma_p+m)$ is a constant at a fixed amount, maximizing on the first branch is maximizing φ_J and on the second φ_{J+m}. Differentiating (2) gives $\varphi_J'(\pi) = -\Phi(\pi; J)/\pi^2$, so Lemma 7 makes φ_J strictly increasing below π_1 and strictly decreasing above it, and φ_{J+m} strictly increasing below π_2 and strictly decreasing above it. Each branch therefore has a unique maximizer, the root delivered by Lemma 5 for that branch.

Step 3: which branch wins. By Step 1 the low-price branch is the set $\pi_{\min} \ge \widehat{V}_p$ and the high-price branch the set $\pi_{\min} \le \widehat{V}_p$, and by (6),

$$\sigma_p \le T_1 \iff \pi_1 \ge \widehat{V}_p, \qquad \sigma_p \le T_2 \iff \pi_2 \ge \widehat{V}_p,$$

since $\pi_i \geq \widehat{V}_p = (J+m+\sigma_p)/k$ reads $\sigma_p \leq k\pi_i - (J+m)$. Take the three cases in turn, using throughout that the two branches agree at $\widehat{V}_p$ and that φ_J and φ_{J+m} are strictly quasi-concave with peaks at π_1 and π_2.

If $\sigma_p \leq T_1$, then $\widehat{V}_p \leq \pi_1 < \pi_2$. The low-price branch contains its own peak and attains $\varphi_J(\pi_1)$; the high-price branch lies entirely below π_2, where φ_{J+m} is strictly increasing, so it is maximized at its endpoint $\widehat{V}_p$, which the low-price branch also attains. The optimum is $\pi_{\min} = \pi_1$.

If $T_1 < \sigma_p \leq T_2$, then $\pi_1 < \widehat{V}_p \leq \pi_2$. The low-price branch lies entirely above π_1, where φ_J is strictly decreasing, and the high-price branch entirely below π_2, where φ_{J+m} is strictly increasing: both are maximized at the point they share, and the optimum is $\pi_{\min} = \widehat{V}_p$. Then $V = \widehat{V}_p = V_p$ by Step 1.

If $\sigma_p > T_2$, then $\pi_1 < \pi_2 < \widehat{V}_p$. The low-price branch is again maximized at $\widehat{V}_p$, while the high-price branch contains its own peak, so it attains $\varphi_{J+m}(\pi_2) > \varphi_{J+m}(\widehat{V}_p)$, which is the common value at the seam. The optimum is $\pi_{\min} = \pi_2$.

Step 4: the price. In each case the capitalization table binds and $V = \pi_{\min}$ (Proposition 1). In the first, $(\sigma_p+m)/V_p^* = k - J/\pi_1$; in the third, $\sigma_p/V_p^* = k-(J+m)/\pi_2$; in the second, $V_p^* = \widehat{V}_p$ by construction. Inverting gives the three displayed prices. $\square$

Proof of Proposition 10

Proof. Write $k := 1-\rho_a^0$, and recall $W(J) = \varphi_J(\pi(J))$ from (2) and $M(x) \gtreqless x$ according as $x \lesseqgtr \bar{x}$, since $M(x) - x$ is strictly decreasing with its zero at $\bar{x}$ (proof of Proposition 3). By Lemma 2, $\pi(J^\dagger) = \bar{x}$ and $\pi(\cdot)$ is strictly increasing, so $\bar{x} \leq \pi_1$, $\pi_1 < \bar{x} \leq \pi_2$ and $\pi_2 < \bar{x}$ are the three cases $J \geq J^\dagger$, $J < J^\dagger \leq J+m$ and $J+m < J^\dagger$.

Participation regime by regime. Take the optimum of Proposition 9 in each of its three ranges and ask when its value is non-negative. Write $A_1 := k - J/\pi_1$ and $A_2 := k-(J+m)/\pi_2$, both positive by Lemma 2, so that $W(J) = A_1 M(\pi_1)$, $W(J+m) = A_2 M(\pi_2)$, and

$$T_1 \;=\; A_1\pi_1 - m, \qquad T_2 \;=\; A_2\pi_2. \tag{9}$$

On $\sigma_p \leq T_1$ the payoff is $\frac{\sigma_p}{\sigma_p+m}W(J) - \sigma_p$, non-negative if and only if $\sigma_p + m \leq W(J)$. On $T_1 < \sigma_p \leq T_2$ it is $\sigma_p\big(M(\widehat{V}_p)/\widehat{V}_p - 1\big)$, non-negative if and only if $\widehat{V}_p \leq \bar{x}$, that is $\sigma_p \leq k\bar{x} - (J+m)$; and $\widehat{V}_p = (J+m+\sigma_p)/k$ runs over $[\pi_1,\pi_2]$ as σ_p runs over $[T_1,T_2]$. On $\sigma_p > T_2$ it is $W(J+m) - \sigma_p$, non-negative if and only if $\sigma_p \leq W(J+m)$.

The three cases. Suppose $\bar{x} \leq \pi_1$. Then $M(\pi_1) \leq \pi_1$, so by (9) the first cap $W(J) - m = A_1 M(\pi_1) - m$ is at most $A_1\pi_1 - m = T_1$: the first range is cut short by participation. The second contributes nothing, since $\widehat{V}_p \geq \pi_1 \geq \bar{x}$ there; and neither does the third, since $\pi_2 > \pi_1 \geq \bar{x}$ gives $M(\pi_2) < \pi_2$ and hence $W(J+m) < T_2$ by (9). The cap is $W(J) - m$ and only the first regime is reached.

Suppose $\pi_1 < \bar{x} \le \pi_2$. Now $M(\pi_1) > \pi_1$ gives $W(J) - m > T_1$, so every amount in the first range participates. In the second, participation holds up to $k\bar{x} - (J+m)$, which lies in $(T_1, T_2]$ because $\pi_1 < \bar{x} \le \pi_2$. The third is again empty, since $M(\pi_2) \le \pi_2$. The cap is $k\bar{x} - (J+m)$ and the first two regimes are reached.

Suppose $\pi_2 < \bar{x}$. The first range participates throughout as before; so does the second, since $\widehat{V}_p \le \pi_2 < \bar{x}$ on it; and in the third, participation holds up to $W(J+m) = A_2 M(\pi_2) > A_2\pi_2 = T_2$. The cap is $W(J+m)$ and all three regimes are reached.

In each case the amounts a participating investor advances form the single interval $[0, \bar{\sigma}_p^m]$ claimed. □

Proof of Corollary 2

Proof. If $J \ge J^\dagger$ then $\bar{x} \le \pi_1$, and by Proposition 10 every participating amount satisfies $\sigma_p \le W(J) - m \le T_1$: the first regime of Proposition 9 is played, with $V_p^* = (\sigma_p + m)/(k - J/\pi_1)$ and $\pi^*_{\min} = \pi_1$. That threshold is the root of $\Phi(\cdot\,; J)$, which is the equation of Corollary 1 with ρ_a replaced by ρ_a^0, and the seed investor holds $J/V = J/\pi_1$, its benchmark stake. For the strict inequality,

$$V_p^* \;<\; V = \pi_1 \quad\iff\quad \sigma_p + m \;<\; k\,\pi_1 - J \;=\; A_1\pi_1 \quad\iff\quad \sigma_p \;<\; T_1,$$

and participation gives $\sigma_p + m \le A_1 M(\pi_1)$ with $M(\pi_1) \le \pi_1$, the second inequality strict unless $\pi_1 = \bar{x}$. Finally the clause converts at V_p^*, so the accelerator holds $\rho_a^* = \rho_a^0 + m/V_p^*$ and the pre-seed investor holds

$$\rho_p^* \;=\; \frac{\sigma_p}{V_p^*} \;=\; \left(1 - \rho_a^0 - \frac{J}{\pi_1}\right) - \frac{m}{V_p^*},$$

its benchmark share net of the accelerator's gain. □

Proof of Remark 2

Proof. A best-price claim converts at $\sigma_p/\min(V, V_p)$, so it differs from the fixed-base claim priced at V_p only in states where $V < V_p$. On $\sigma_p \le T_1$ there are none: by Proposition 11(i) the cap binds and the claim converts at $V_p^* = (\sigma_p + m)/(k - J/\pi_1)$, which satisfies $V_p^* \le V = \pi_1$ because $\sigma_p + m \le k\pi_1 - J$ is $\sigma_p \le T_1$. Hence $\min(V, V_p^*) = V_p^*$: the two instruments induce the same conversion, the same capitalization table and the same payoff, so the investor's problem—and therefore its optimal price—is unchanged. □

Proof of Proposition 11

Proof. Write $k := 1 - \rho_a^0$. Both claims now convert at $\min(V, V_p)$, so the two branches of Step 1 in the proof of Proposition 9 change in one place. On $V_p \le \widehat{V}_p$ the minimum is V_p for both, the capitalization table is $\rho_a^0 + (\sigma_p + m)/V_p + J/V = 1$ as before, and the investor's problem is the low-price branch unchanged; by Step 3 its optimum is π_1 when $\sigma_p \le T_1$, and the endpoint $\widehat{V}_p$ otherwise.

On $V_p \ge \widehat{V}_p$ the minimum is V for both, so the table reads $\rho_a^0 + (\sigma_p + m)/V + J/V = 1$, that is $V = (J + m + \sigma_p)/k = \widehat{V}_p$, independently of V_p. Price and threshold are then constants and $\Pi_p = \sigma_p\big(M(\widehat{V}_p)/\widehat{V}_p - 1\big)$ for every $V_p \ge \widehat{V}_p$: the whole range is optimal and delivers one payoff, the value the low-price branch attains at its endpoint. Hence for $\sigma_p \le T_1$ the optimum is π_1, and for $\sigma_p > T_1$ it is $\widehat{V}_p$. □

Proof of Proposition 12

Proof. At any amount, the fixed-base claim attains the larger of the two branch maxima of Proposition 9, while by Proposition 11 the best-price claim attains the larger of the low-price maximum and the plateau value $\sigma_p\big(M(\widehat{V}_p)/\widehat{V}_p - 1\big)$, which is the value both branches take at $\widehat{V}_p$. Since the high-price branch is maximized at π_2 and $\varphi_{J+m}(\pi_2) \ge \varphi_{J+m}(\widehat{V}_p)$, the fixed-base claim is worth weakly more; and the inequality is strict exactly when $\pi_2 < \widehat{V}_p$, that is when $\sigma_p > T_2$.

For the last statement, suppose $\pi_1 \ge \bar{x}$. By Proposition 10 no participating amount exceeds T_1, a fortiori none exceeds T_2, so a fixed-base holder advancing one would earn a strictly negative payoff and not propose; both claims are then worth zero. A strict gap at an amount the investor would actually advance therefore requires $\pi_1 < \bar{x}$. □

Proof of Corollary 3

Proof. By Proposition 3 the pivot $\tilde{\Delta}$ is the zero of $h(\Delta) := \Pi_p(\Delta) - \Pi_p(0)$ on the increasing branch, and the same construction applies under the clause to $h^m(\Delta) := \Pi_p^m(\Delta) - \Pi_p^m(0)$. By Proposition 13, $\partial\Pi_p^m/\partial\Delta \ge \partial\Pi_p/\partial\Delta$ at every Δ, with strict inequality wherever $V_p^* < V$. Integrating from 0,

$$h^m(\Delta) \;=\; \int_0^\Delta \frac{\partial \Pi_p^m}{\partial x}\,dx \;\ge\; \int_0^\Delta \frac{\partial \Pi_p}{\partial x}\,dx \;=\; h(\Delta) \qquad \text{for every } \Delta > 0.$$

Both functions are strictly increasing on that branch (Lemma 3), so the zero of the larger one comes first: $\tilde{\Delta}^m \le \tilde{\Delta}$, strictly when the inequality between the integrands is strict on a set of positive measure. Since $\sigma_p^* = \beta$ exactly when $\beta > b + \tilde{\Delta}$, lowering the pivot enlarges the set of budgets on which the investor pre-funds. □

Proof of Proposition 13

Proof. In regime (i) the pre-seed investor and the MFN SAFE convert on the same base, so the investor holds the fraction $\sigma_p/(\sigma_p+m)$ of a total claim whose value is $W(J)$, and $\Pi_p = \frac{\sigma_p}{\sigma_p+m} W(J) - \sigma_p$ with $\sigma_p = b+\Delta$ and $J = I-\Delta$. Differentiating, and using $W'(J) = -\Lambda(J)$ from Lemma 3,

$$\frac{\partial \Pi_p}{\partial \Delta} = \frac{m}{(\sigma_p+m)^2} W(J) + \frac{\sigma_p}{\sigma_p+m} \Lambda(J) - 1.$$

Subtracting $\Lambda(J)-1$ leaves $\frac{m}{(\sigma_p+m)^2}\big[W(J) - (\sigma_p+m)\Lambda(J)\big]$. Substituting $W(J) = \rho_p^{\text{tot}} M(\pi_1)$ with $\rho_p^{\text{tot}} = (\sigma_p+m)/V_p^*$ and $\Lambda(J) = M(\pi_1)/\pi_1$ collapses the bracket to $(\sigma_p+m)M(\pi_1)\big[1/V_p^* - 1/\pi_1\big]$, which is non-negative because the first regime carries $V_p^* \le V = \pi_1$ by Proposition 9, and strictly positive when that inequality is strict. $\square$

Proof of Proposition 14

Proof. The two prices generate the table. In regime (i), $V_p^* < V = \pi_1$, so $P^* = V_p^* = (\sigma_p+m)/(k - J/\pi_1)$ and $m/P^* = \big(k - J/\pi_1\big)m/(\sigma_p+m)$, which is the accelerator's entry; the seed entry is $J/\pi_1 = J/V^*$. In regime (ii) the two prices coincide at $\widehat{V}_p$ and the entries are $m/\widehat{V}_p$ and $J/\widehat{V}_p$. In regime (iii), $V_p^* > V = \pi_2$, so $P^* = \pi_2$ and the entries are m/π_2 and J/π_2. In each case the three stakes of Proposition 9 sum to k, which gives the third expression in (7), and the fourth is the definition of the pre-seed price. Setting $m = 0$ leaves $\rho_p = k - J/\pi_1$ and $V_p = \sigma_p/\rho_p$, the benchmark column.

Continuity and monotonicity. By (6), $\sigma_p = T_1$ gives $\widehat{V}_p = (J+m+T_1)/k = \pi_1$, and $k - J/\pi_1 = (k\pi_1 - J)/\pi_1 = (T_1+m)/\pi_1$, so the regime (i) price evaluated at T_1 is $(T_1+m)/\big((T_1+m)/\pi_1\big) = \pi_1$ as well: both prices are continuous at T_1 and equal π_1 there. Likewise $\sigma_p = T_2$ gives $\widehat{V}_p = \pi_2$ and $k - (J+m)/\pi_2 = T_2/\pi_2$, so the regime (iii) price evaluated at T_2 is π_2: both are continuous at T_2 and equal π_2. Between the joins, P^* is affine and strictly increasing in σ_p on (i) and on (ii) and constant on (iii), while V^* is constant at π_1 on (i), equal to $\widehat{V}_p$ and strictly increasing on (ii), and constant at π_2 on (iii); $\pi_1 < \pi_2$ by Lemma 2. Continuity of the entries follows from (7).

Across the regimes. m/P^* and J/V^* are non-increasing in σ_p by the previous step, so ρ_a^* and ρ_s^* fall and $\rho_p^* = k - J/V^* - m/P^*$ rises. The pre-seed price σ_p/ρ_p^* is affine and strictly increasing in σ_p within each regime and continuous at the joins, hence increasing throughout. $V^* = \pi^*_{\min}$ rises by the previous step and $1 - F$ is decreasing.

Against the benchmark. In regime (i), $\rho_p^* = \rho_p^{\text{bench}}\,\sigma_p/(\sigma_p+m) < \rho_p^{\text{bench}}$. On (ii), $\widehat{V}_p \le \pi_2$ gives $\rho_p^* = k - (J+m)/\widehat{V}_p \le k - (J+m)/\pi_2$, the regime (iii) value, so it suffices to compare that value with the benchmark: $k - (J+m)/\pi(J+m) < k - J/\pi(J)$ holds if and only if $J \mapsto J/\pi(J)$ is strictly increasing. Writing $J = \psi(\pi)$ with $\pi = \pi(J)$

increasing (Lemma 2), $J/\pi(J) = \psi(\pi)/\pi = k\pi^2 f/\big(\pi^2 f + M\big)$, which increases in π if and only if $M/(\pi^2 f)$ decreases. Using $M' = -\pi f$, that derivative has the sign of $-\big[\pi^2 f^2 + M(2f + \pi f')\big]$, so the claim is

$$\pi^2 f(\pi) \; + \; M(\pi)\big(2 - c\,\pi\big) \; > \; 0, \qquad c := -f'(\pi)/f(\pi),$$

immediate when $c \le 2/\pi$ since $f > 0$. When $c > 2/\pi$ the second term is negative, and the log-concavity bound $M(\pi) \le f(\pi)(c\pi + 1)/c^2$ established in the proof of Lemma 2 can only lower the left-hand side:

$$\pi^2 f + M\big(2 - c\pi\big) \; \ge \; \frac{f}{c^2}\Big(c^2\pi^2 + (c\pi + 1)(2 - c\pi)\Big) \; = \; \frac{f}{c^2}\big(c\pi + 2\big) \; > \; 0.$$

Hence $\rho_p^* < \rho_p^{\text{bench}}$ in all three regimes, and $V_p^* = \sigma_p/\rho_p^* > \sigma_p/\rho_p^{\text{bench}}$ reverses the inequality. For the remaining rows, $V^* = \pi_1$ in regime (i) leaves ρ_s^*, the round price and $1 - F(\pi^*_{\min})$ at their benchmark values, while $V^* > \pi_1$ strictly on (ii) and (iii)—by $\sigma_p > T_1$ and $\pi_1 < \pi_2$ respectively—raises the threshold, lowers J/V^* and lowers $1 - F(V^*)$. $\square$